

2003

**HIGHER SCHOOL CERTIFICATE
EXAMINATION**

Chemistry

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Draw diagrams using pencil
- Board-approved calculators may be used
- A data sheet and a Periodic Table are provided at the back of this paper
- Write your Centre Number and Student Number at the top of pages 13, 17, 21 and 25

Total marks – 100

Section I Pages 2–28

75 marks

This section has two parts, Part A and Part B

Part A – 15 marks

- Attempt Questions 1–15
- Allow about 30 minutes for this part

Part B – 60 marks

- Attempt Questions 16–29
- Allow about 1 hour and 45 minutes for this part

Section II Pages 29–37

25 marks

- Attempt ONE question from Questions 30–34
- Allow about 45 minutes for this section

Section I

75 marks

Part A – 15 marks

Attempt Questions 1–15

Allow about 30 minutes for this part

Use the multiple-choice answer sheet.

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
 A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

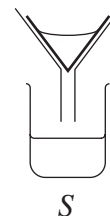
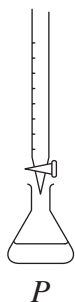
A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word *correct* and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐
 correct
 ↙

- 1** Which of the following is an acid–base indicator?
- (A) Methanol
 - (B) Methyl orange
 - (C) Methanoic acid
 - (D) Methyl ethanoate
- 2** Which of the following is a transuranic element?
- (A) Caesium
 - (B) Cerium
 - (C) Chromium
 - (D) Curium
- 3** Which instrument is used to detect radiation from radioactive isotopes?
- (A) pH meter
 - (B) Geiger counter
 - (C) Ion-selective electrode
 - (D) Atomic absorption spectrophotometer (AAS)
- 4** In which layer of the atmosphere does ozone act as a UV radiation shield?
- (A) Mesosphere
 - (B) Stratosphere
 - (C) Thermosphere
 - (D) Troposphere

- 5 Which of the following could be used to determine the total dissolved solids in a sample of muddy river water?



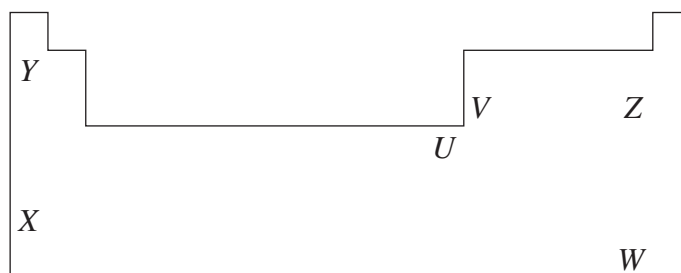
- (A) *P* and *Q*
(B) *R* and *S*
(C) *P* and *R*
(D) *Q* and *S*
- 6 The table gives the heat of combustion in kJ g^{-1} for a number of different fuels.

<i>Fuel</i>	<i>Heat of combustion</i> (kJ g^{-1})
Methanol	22.7
Ethanol	29.6
Propanol	33.6
Petrol (octane)	47.8

The heat of combustion in kJ mol^{-1} for one of the fuels was calculated as 2016 kJ mol^{-1} . What was the fuel?

- (A) Methanol
(B) Ethanol
(C) Propanol
(D) Petrol

- 7 The diagram is a representation of the Periodic Table. The positions of six different elements are shown.



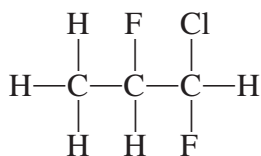
What are the reactions of oxides of these elements with acid and with base?

	<i>Oxide reacts with acid</i>	<i>Oxide reacts with base</i>	<i>Oxide reacts with acid and with base</i>
(A)	Z	X	V
(B)	Y	X	U
(C)	X	Z	V
(D)	V	W	Y

- 8 A sulfuric acid solution has a concentration of $5 \times 10^{-4} \text{ mol L}^{-1}$.

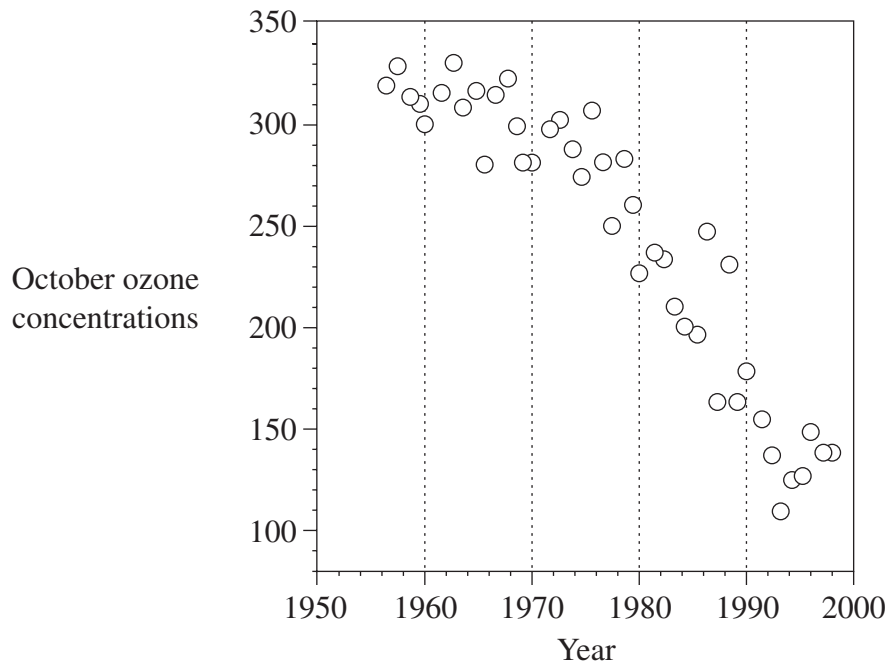
What is the pH of this solution, assuming the acid is completely ionised?

- (A) 3.0
 (B) 3.3
 (C) 3.6
 (D) 4.0
- 9 What is the name of the compound shown?



- (A) 1-chloro-1,2-difluoropropane
 (B) 3-chloro-2,3-difluoropropane
 (C) 1,2-difluoro-1-chloropropane
 (D) 1-chloro-1,2-difluoropentane

- 10 The graph shows October ozone concentrations above Halley Bay in Antarctica between 1956 and 1998.

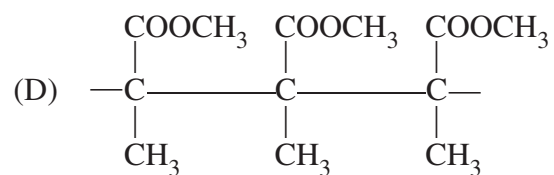
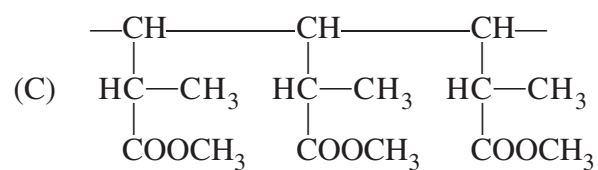
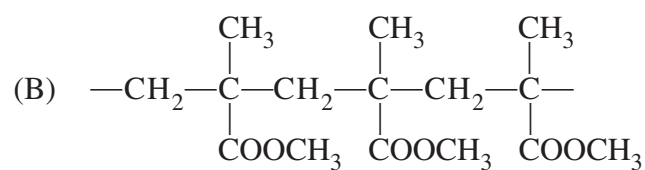
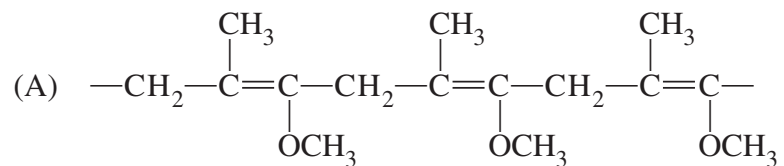
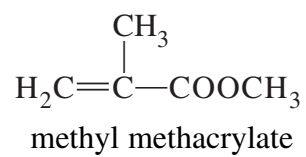


Reproduced with the permission of Oxford University Press, London

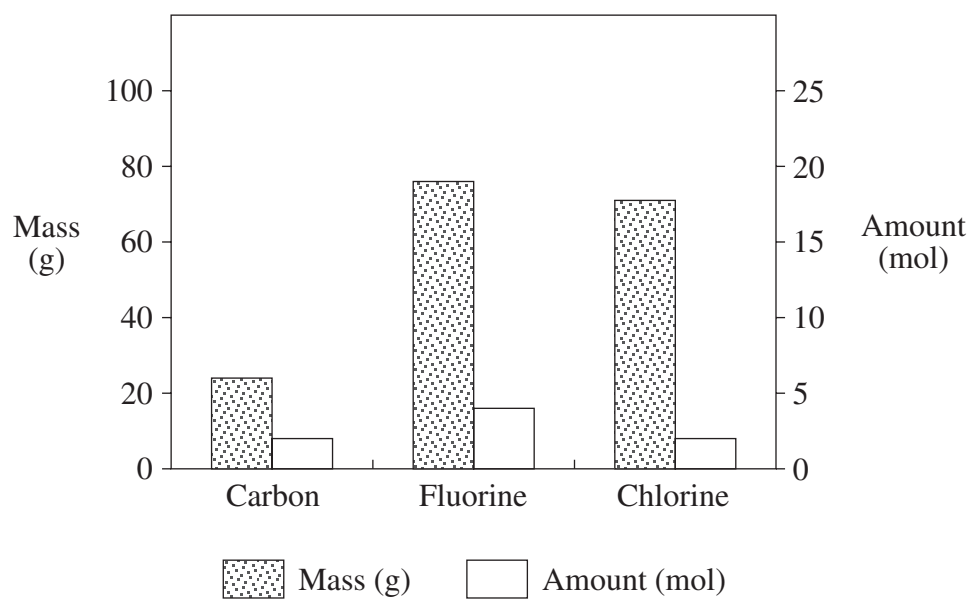
Based on these data alone, which of the following is a valid statement about the concentration of ozone above Halley Bay?

- (A) It was greater in 1998 than in 1993.
- (B) It will be greater in 2004 than in 1998.
- (C) The variation in ozone concentration between 1960 and 1970 was due to changes in atmospheric CFC concentrations.
- (D) The variation in ozone concentration from one year to the next is due only to changes in atmospheric CFC concentrations.

11 Which polymer is made by the polymerisation of methyl methacrylate?



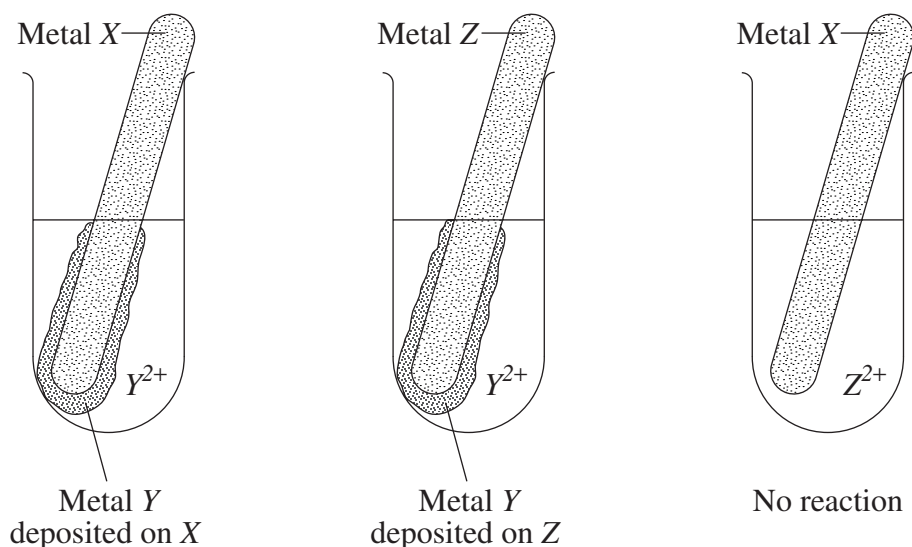
- 12 The graph shows the mass and amount of carbon, fluorine and chlorine atoms in one mole of a compound.



What is the molecular formula for this compound?

- (A) CF_2Cl
- (B) CF_2Cl_2
- (C) $\text{C}_2\text{F}_3\text{Cl}_3$
- (D) $\text{C}_2\text{F}_4\text{Cl}_2$

- 13** A student performed three tests to investigate the relative activity of metals. In each test a metal strip was placed in a solution containing ions of a different metal. The results are shown in the diagrams.



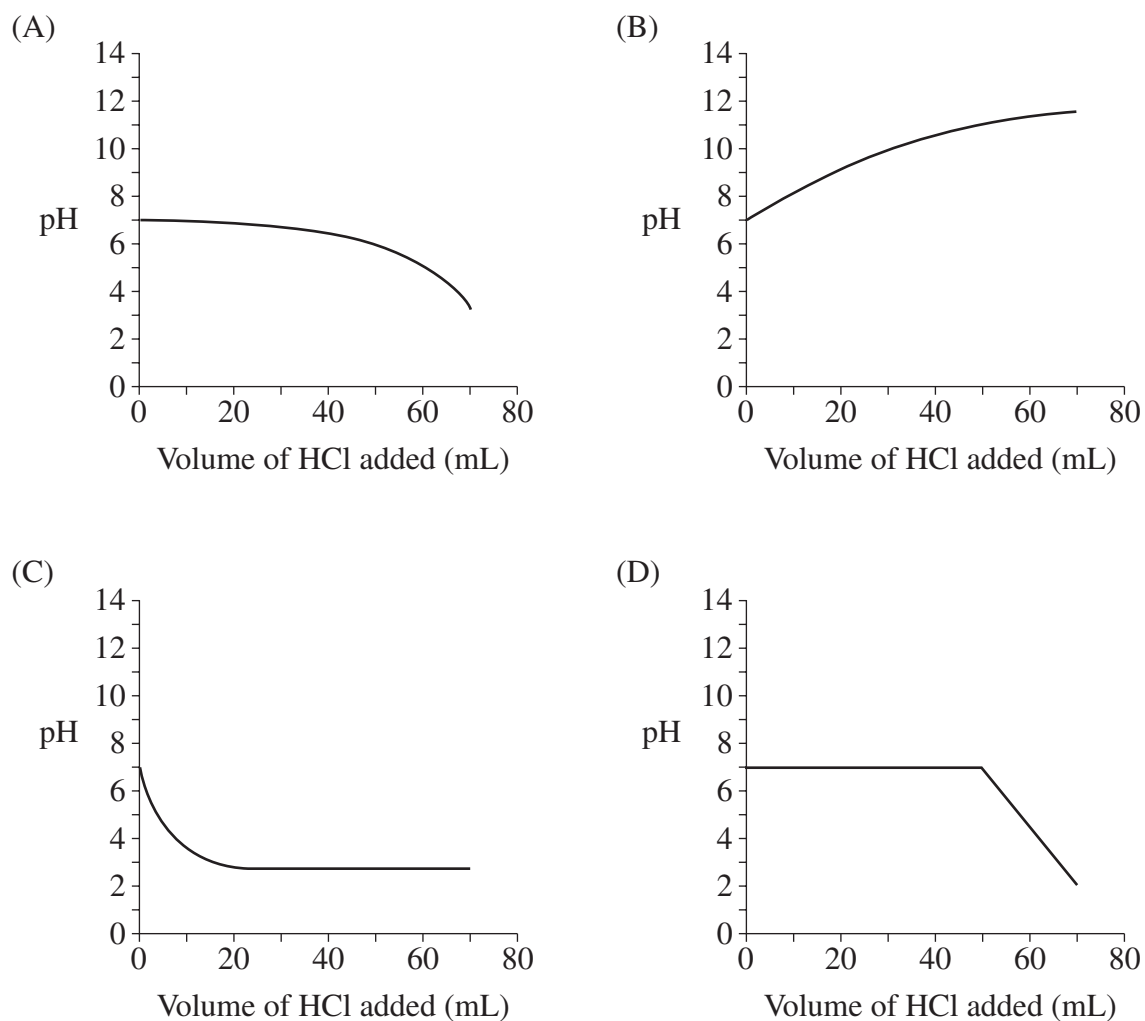
What is the order of activity of the metals, based on these results?

- (A) $X > Z > Y$
(B) $Y > X > Z$
(C) $Z > Y > X$
(D) $Z > X > Y$
- 14** In a titration of a strong base with a strong acid, the following procedure was used:
1. A burette was rinsed with water and then filled with the standard acid.
 2. A pipette was rinsed with some base solution.
 3. A conical flask was rinsed with some base solution.
 4. A pipette was used to transfer a measured volume of base solution into the conical flask.
 5. Indicator was added to the base sample and it was titrated to the endpoint with the acid.

Which statement is correct?

- (A) The calculated base concentration will be correct.
(B) The calculated base concentration will be too low.
(C) The calculated base concentration will be too high.
(D) No definite conclusion can be reached about the base concentration.

- 15 Which of the following graphs shows how pH will vary when dilute HCl is added to 100 mL of dilute natural buffer solution with an initial pH of 7.0?



BLANK PAGE

BLANK PAGE

--	--	--	--	--

Centre Number

Section I (continued)

--	--	--	--	--	--	--	--	--

Student Number

Part B – 60 marks

Attempt Questions 16–29

Allow about 1 hour and 45 minutes for this part

Answer the questions in the spaces provided.

Show all relevant working in questions involving calculations.

Marks

Question 16 (3 marks)

You performed a first-hand investigation that monitored mass changes during the fermentation of glucose to ethanol.

- (a) Outline the procedure you used.

2

.....

.....

.....

.....

.....

.....

.....

.....

.....

- (b) Write a balanced chemical equation for this reaction.

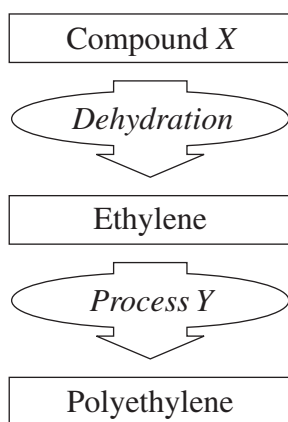
1

.....

.....

Question 17 (5 marks)

The flowchart shows the production of polyethylene.



- (a) Identify Compound X.

1

.....

- (b) Describe *Process Y*.

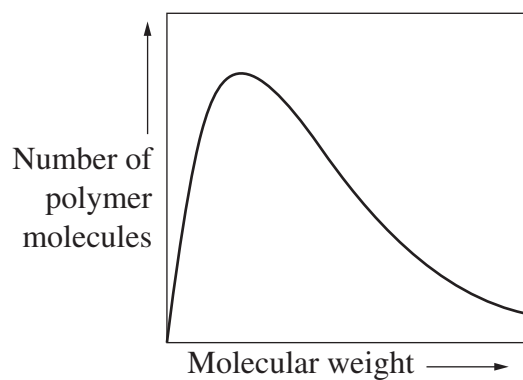
3

.....
.....
.....
.....
.....
.....
.....
.....

Question 17 continues on page 15

Question 17 (continued)

A sample of polyethylene was produced by *Process Y*. The following graph shows the distribution of molecular weights of polymer molecules in the sample.



- (c) Why is a range of molecular weights observed?

1

.....

.....

.....

.....

End of Question 17**Please turn over**

Question 18 (4 marks)

Describe how commercial radioisotopes are produced, and how transuranic elements are produced.

4

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

--	--	--	--	--

Centre Number

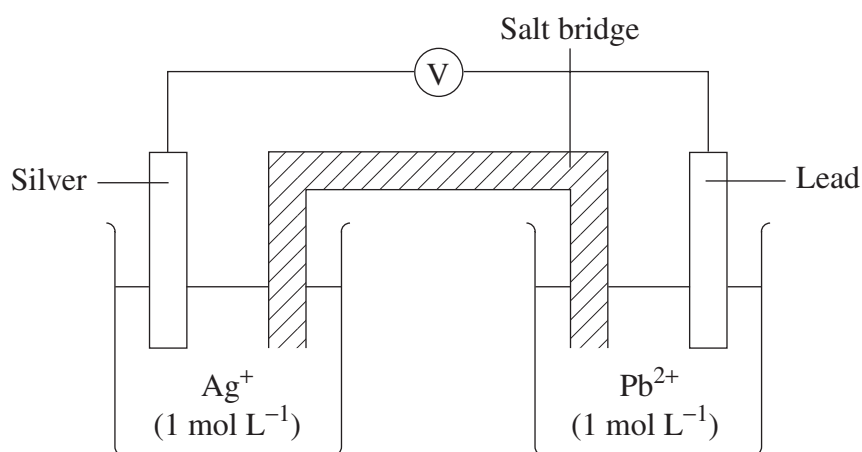
Section I — Part B (continued)

--	--	--	--	--	--	--	--	--

Student Number

Marks

Question 19 (3 marks)



- (a) Identify the cathode in this diagram.

1

.....

- (b) Write the net redox equation for the cell reaction, and calculate the cell potential (
- $E^\ominus$
-).

2

.....

.....

.....

.....

.....

.....

Marks

Question 20 (5 marks)

Assess the suitability of biomass as a future source of energy and chemicals for industry.

5

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and extend across the width of the page. There are no margins, text, or other markings on the paper.

Question 21 (5 marks)

You performed a first-hand investigation to prepare an ester by reflux.

- (a) Identify the products formed when propanoic acid and butanol are refluxed with acid catalyst. **1**

.....

.....

- (b) Draw a fully labelled diagram of the equipment assembled for use. **2**

- (c) Outline the advantages of using reflux to prepare the ester. **2**

.....

.....

.....

.....

.....

.....

Question 22 (3 marks)

- (a) Write a balanced chemical equation for the complete combustion of ethanol. **1**

.....

.....

- (b) A mass of 72.5 g of ethanol was burnt completely in air. Calculate the volume of carbon dioxide that was produced at 25°C and 100 kPa. **2**

.....

.....

.....

.....

.....

.....

--	--	--	--	--

Centre Number

Section I — Part B (continued)

--	--	--	--	--	--	--	--	--

Student Number

Marks

Question 23 (4 marks)

25.0 mL of 0.12 mol L^{-1} standard barium hydroxide solution was titrated with nitric acid. The results are recorded in the table.

<i>Titration</i>	<i>Volume of nitric acid used (mL)</i>
1	20.4
2	18.1
3	18.2
4	18.1

- (a) Write a balanced chemical equation for the reaction of barium hydroxide with nitric acid. 1

.....

.....

- (b) Calculate the concentration of the nitric acid. 3

.....

.....

.....

.....

.....

.....

.....

.....

Question 24 (4 marks)

Discuss factors that must be considered when using neutralisation reactions to safely minimise damage in chemical spills.

4

.....

.....

.....

.....

.....

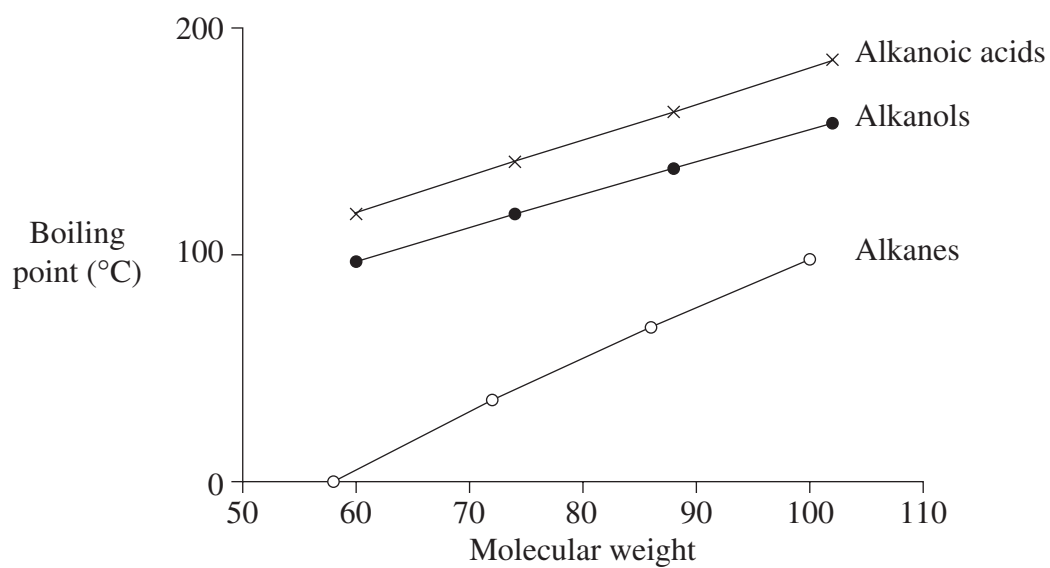
.....

.....

.....

.....

4

This image shows a full page of white paper with horizontal dashed lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the page.

Question 26 (4 marks)

Describe the process of eutrophication, and assess the suitability of water quality tests used to monitor it.

4

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

--	--	--	--	--

Centre Number

Section I — Part B (continued)

--	--	--	--	--	--	--	--	--

Student Number

Marks

Question 27 (5 marks)

A student carried out an investigation to analyse the sulfate content of lawn fertiliser. The student weighed out 1.0 g of fertiliser and dissolved it in water. 50 mL of 0.25 mol L⁻¹ barium chloride solution was then added. A white precipitate of barium sulfate formed, which weighed 1.8 g.

- (a) Calculate the percentage by mass of sulfate in the fertiliser.

2

.....

.....

.....

.....

.....

.....

- (b) Evaluate the reliability of the experimental procedure used.

3

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

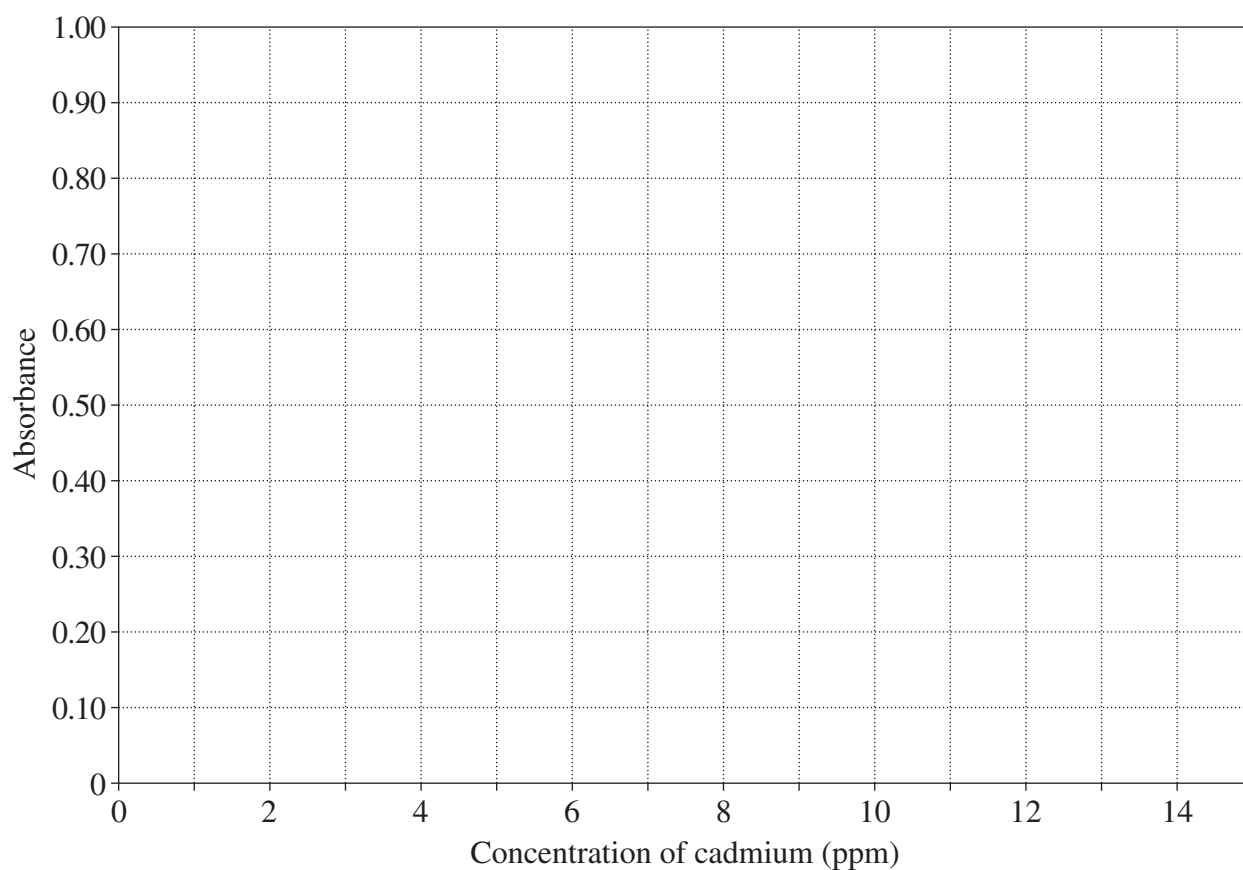
.....

Question 28 (4 marks)

The results of analysis of a set of standard cadmium solutions are presented in the table.

<i>Concentration of cadmium standard solution (ppm)</i>	<i>Absorbance</i>
0	0.00
3	0.22
6	0.38
9	0.62
12	0.83

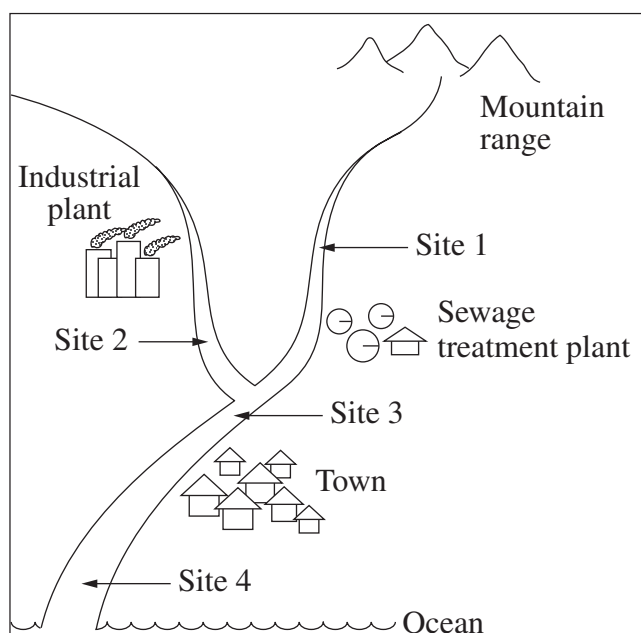
(a) Draw an appropriate graph of the data.

2

Question 28 continues on page 27

Question 28 (continued)

The map shows a catchment area. There is an industrial plant, a sewage treatment plant and a small town, all of which discharge water into the river. Water samples were collected at four sites.



The results of analysis of cadmium levels from these four sites are given in the table.

<i>Sample site</i>	<i>Absorbance</i>
Site 1	0.08
Site 2	0.15
Site 3	0.55
Site 4	0.40

- (b) Justify your conclusion about the most likely source of cadmium pollution.

2

.....

.....

.....

.....

.....

.....

.....

End of Question 28

7

This image shows a full page of white paper with horizontal dotted lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the page.

Chemistry

Section II

25 marks

Attempt ONE question from Questions 30–34

Allow about 45 minutes for this section

Answer the question in a writing booklet. Extra writing booklets are available.

Show all relevant working in questions involving calculations.

	Pages
Question 30 Industrial Chemistry	30
Question 31 Shipwrecks, Corrosion and Conservation	31
Question 32 The Biochemistry of Movement	32–33
Question 33 The Chemistry of Art	34–35
Question 34 Forensic Chemistry	36–37

Question 30 — Industrial Chemistry (25 marks)

- (a) (i) Identify ONE use of sulfuric acid in industry. **1**
- (ii) One of the starting materials used for preparing sulfuric acid is sulfur. Describe the process used to extract sulfur from mineral deposits. **3**
- (b) During your practical work you performed a first-hand investigation to identify the products of electrolysis of sodium chloride.
- (i) Describe ONE precaution you took to minimise hazards, or to dispose of reactants and products safely. **1**
- (ii) Outline the procedure you used to identify the products of electrolysis of sodium chloride. **3**
- (c) Analyse how an understanding of the structure and cleaning action of soaps led to the development of synthetic detergents. **5**
- (d) The Ostwald process is used for making nitric acid from ammonia, and involves several equilibrium steps.
- (i) Identify the only factor that changes the value of an equilibrium constant. **1**
- (ii) One step in the process produces nitrogen dioxide according to the equation:
- $$2\text{NO}(g) + \text{O}_2(g) \rightleftharpoons 2\text{NO}_2(g).$$
- This reaction is exothermic. Describe TWO methods that could be used to increase the yield of nitrogen dioxide.
- (iii) A 1 L reaction vessel initially contained 0.25 mol NO and 0.12 mol O₂. After equilibrium was established there was only 0.05 mol NO. **3**
- Calculate the equilibrium constant for the reaction. Show all relevant working.
- (e) Assess how environmental issues have been addressed in an industrial method of production of an acid, and an industrial method of production of a base. **6**

Question 31 — Shipwrecks, Corrosion and Conservation (25 marks)

- | | | |
|---|---|----------|
| (a) | (i) Identify ONE passivating metal. | 1 |
| | (ii) Account for the differences in corrosion of active and passivating metals. Include a relevant balanced chemical equation in your answer. | 3 |
| (b) During your practical work you performed a first-hand investigation to identify the factors that affect the rate of an electrolysis reaction. | | |
| | (i) Describe ONE precaution you took to minimise hazards, or to dispose of reactants and products safely. | 1 |
| | (ii) Outline the procedure you used to show how ONE factor affects the rate of an electrolysis reaction. | 3 |
| (c) The Titanic struck an iceberg in 1912 and sank to a depth of more than three kilometres. | | |
| Analyse how theories about corrosion at great ocean depth have changed since the recent discovery of extensive corrosion on wrecks such as the Titanic. | | |
| (d) Fishermen face a problem of limiting corrosion of their steel fish hooks. A fisherman has the choice of storing his steel fish hooks in a plastic, copper or aluminium container. | | |
| | (i) What is the main metal in steel? | 1 |
| | (ii) Compare the effectiveness of these containers in limiting corrosion of steel fish hooks. | 5 |
| (e) Assess how an understanding of electrolysis has contributed to the development of methods for cleaning and restoring marine artefacts. | | |
| | | 6 |

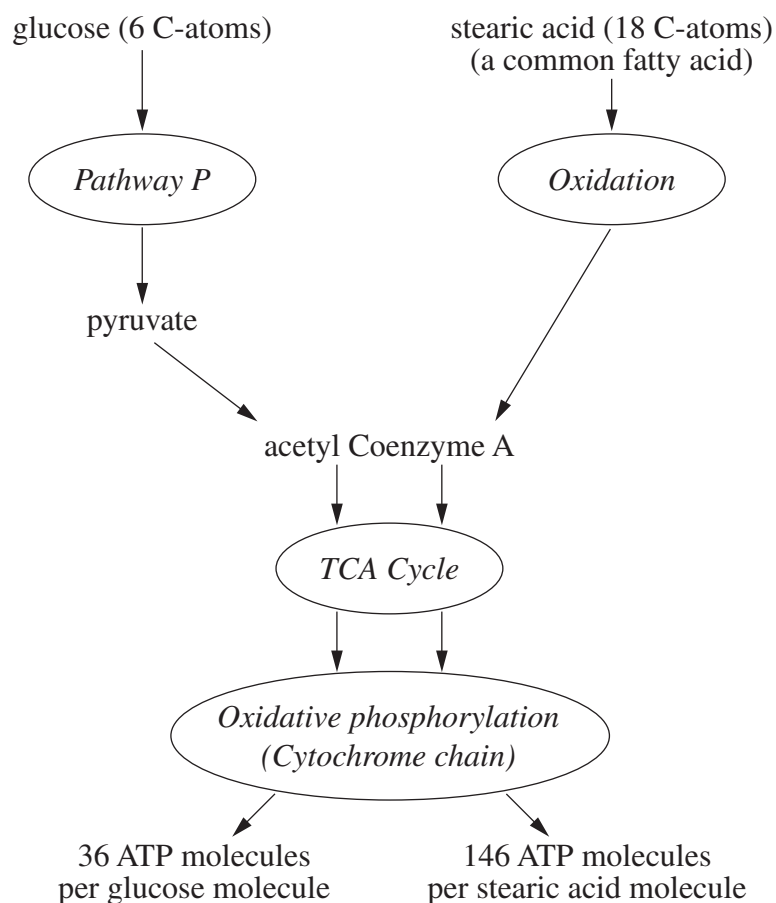
Question 32 — The Biochemistry of Movement (25 marks)

- | | | | |
|--|------|---|----------|
| (a) | (i) | Identify the role of glycogen in human muscle and liver. | 1 |
| | (ii) | Describe the process of bond formation when glucose molecules react to form glycogen. | 3 |
|
(b) During your practical work you performed a first-hand investigation to compare the structures of fatty acids and glycerol. | | | |
| | (i) | Describe the results of your investigation. | 2 |
| | (ii) | Outline why glycerol is more soluble in water than are fatty acids that are commonly found in the body. | 2 |
|
(c) Analyse how an understanding of the composition and structure of proteins led to the current theory of muscle contraction. | | | |
| | | | 5 |

Question 32 continues on page 33

Question 32 (continued)

(d) The flowchart summarises the production of ATP from glucose and stearic acid.

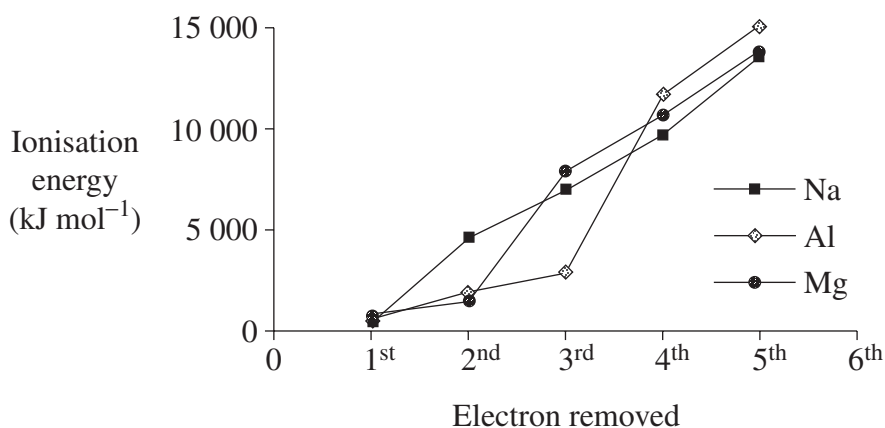


- | | |
|---|----------|
| (i) Identify <i>Pathway P</i> . | 1 |
| (ii) Construct a word-equation that summarises the formation of products in <i>Pathway P</i> , when one molecule of glucose is metabolised. | 2 |
| (iii) Explain why fats can produce more energy per carbon atom than carbohydrates. | 3 |
| | |
| (e) Evaluate the importance of the chemistry of ATP in metabolic processes. | 6 |

End of Question 32

Question 33 — The Chemistry of Art (25 marks)

- (a) (i) What is the maximum number of electrons found in an atomic orbital? 1
- (ii) The graph shows the first five ionisation energies for sodium, aluminium and magnesium. 3



C E Housecroft & E C Constable, 2002, *Chemistry*, 2nd edn, reproduced with permission of Prentice Hall, Harlow, England.

Explain how the data can be used to provide information about the arrangement of electrons around the atoms.

- (b) During your practical work you performed a first-hand investigation to observe the flame colour of different metal ions.
- (i) Explain the precautions you took during your investigation. 2
- (ii) Outline the procedure you used to observe the flame colour of different metal ions. 2
- (c) Analyse the relationship between the chemical composition and properties, including colour, of pigments used in traditional art by Aboriginal people. 5

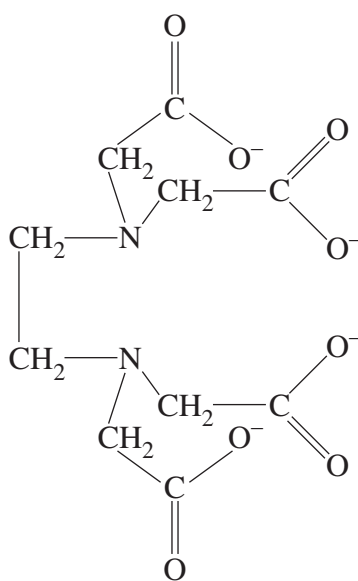
Question 33 continues on page 35

Question 33 (continued)

- (d) In the early 1900s, many Australian children were diagnosed as having Pink disease.

The symptoms included pain, loss of teeth, and the presence of a pink colour on fingers and toes.

The first indication that Pink disease was due to mercury poisoning came from the successful treatment of the children using EDTA.



EDTA

- | | |
|---|----------|
| (i) Identify the block in the Periodic Table in which mercury is present. | 1 |
| (ii) Explain why transition metals such as mercury may have more than one oxidation state. | 2 |
| (iii) Explain why chelating ligands such as EDTA are often used to treat poisoning by heavy metals. | 3 |
| (e) Evaluate the usefulness of the range of technologies used by chemists to identify pigments. | 6 |

End of Question 33

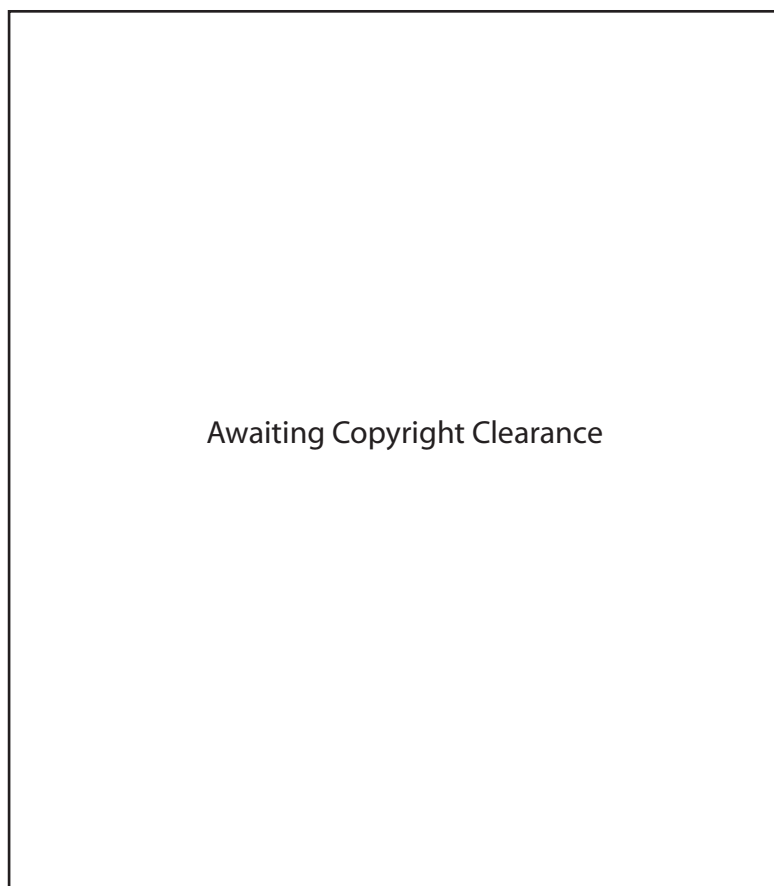
Question 34 — Forensic Chemistry (25 marks)

- (a) (i) Identify the general class of compounds represented by the formula $C_x(H_2O)_y$. **1**
- (ii) Describe TWO tests that can be used to distinguish between some of the following classes of organic compounds: alkanes, alkenes, alkanols and alkanoic acids. **3**
- (b) During your practical work you performed a first-hand investigation to distinguish between reducing and non-reducing sugars.
- (i) Describe ONE precaution you took to minimise hazards, or to dispose of reactants and products safely. **1**
- (ii) Outline the procedure you used to distinguish between reducing and non-reducing sugars. **3**
- (c) Analyse how emission spectra of elements assist in the identification of the origins of a mixture. **5**

Question 34 continues on page 37

Question 34 (continued)

- (d) The diagram shows the results of an investigation to identify the parents of a child. The DNA fingerprints from the mother (M) and child (C) are labelled. Also shown are the DNA fingerprints from two possible fathers (F1) and (F2).



- | | |
|---|----------|
| (i) Identify the more probable father. | 1 |
| (ii) Outline the structure and composition of DNA. | 2 |
| (iii) Describe how DNA fingerprints are produced, and explain why they can be used to show that two people belong to the same family. | 3 |
| (e) Evaluate how the development of chromatographic methods has advanced forensic science. | 6 |

End of paper

BLANK PAGE

BLANK PAGE

BLANK PAGE

DATA SHEET

Avogadro constant, N_A	$6.022 \times 10^{23} \text{ mol}^{-1}$
Volume of 1 mole ideal gas: at 100 kPa and	
at 0°C (273.15 K)	22.71 L
at 25°C (298.15 K)	24.79 L
Ionisation constant for water at 25°C (298.15 K), K_w	1.0×10^{-14}
Specific heat capacity of water	$4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$

Some useful formulae

$$\text{pH} = -\log_{10}[\text{H}^+]$$

$$\Delta H = -m C \Delta T$$

Some standard potentials

$\text{K}^+ + \text{e}^-$	$\rightleftharpoons$	K(s)	-2.94 V
$\text{Ba}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ba(s)	-2.91 V
$\text{Ca}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ca(s)	-2.87 V
$\text{Na}^+ + \text{e}^-$	$\rightleftharpoons$	Na(s)	-2.71 V
$\text{Mg}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Mg(s)	-2.36 V
$\text{Al}^{3+} + 3\text{e}^-$	$\rightleftharpoons$	Al(s)	-1.68 V
$\text{Mn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Mn(s)	-1.18 V
$\text{H}_2\text{O} + \text{e}^-$	$\rightleftharpoons$	$\frac{1}{2}\text{H}_2(\text{g}) + \text{OH}^-$	-0.83 V
$\text{Zn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Zn(s)	-0.76 V
$\text{Fe}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Fe(s)	-0.44 V
$\text{Ni}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ni(s)	-0.24 V
$\text{Sn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Sn(s)	-0.14 V
$\text{Pb}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Pb(s)	-0.13 V
$\text{H}^+ + \text{e}^-$	$\rightleftharpoons$	$\frac{1}{2}\text{H}_2(\text{g})$	0.00 V
$\text{SO}_4^{2-} + 4\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	$\text{SO}_2(\text{aq}) + 2\text{H}_2\text{O}$	0.16 V
$\text{Cu}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Cu(s)	0.34 V
$\frac{1}{2}\text{O}_2(\text{g}) + \text{H}_2\text{O} + 2\text{e}^-$	$\rightleftharpoons$	2OH^-	0.40 V
$\text{Cu}^+ + \text{e}^-$	$\rightleftharpoons$	Cu(s)	0.52 V
$\frac{1}{2}\text{I}_2(\text{s}) + \text{e}^-$	$\rightleftharpoons$	I^-	0.54 V
$\frac{1}{2}\text{I}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	I^-	0.62 V
$\text{Fe}^{3+} + \text{e}^-$	$\rightleftharpoons$	Fe^{2+}	0.77 V
$\text{Ag}^+ + \text{e}^-$	$\rightleftharpoons$	Ag(s)	0.80 V
$\frac{1}{2}\text{Br}_2(\text{l}) + \text{e}^-$	$\rightleftharpoons$	Br^-	1.08 V
$\frac{1}{2}\text{Br}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	Br^-	1.10 V
$\frac{1}{2}\text{O}_2(\text{g}) + 2\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	H_2O	1.23 V
$\frac{1}{2}\text{Cl}_2(\text{g}) + \text{e}^-$	$\rightleftharpoons$	Cl^-	1.36 V
$\frac{1}{2}\text{Cr}_2\text{O}_7^{2-} + 7\text{H}^+ + 3\text{e}^-$	$\rightleftharpoons$	$\text{Cr}^{3+} + \frac{7}{2}\text{H}_2\text{O}$	1.36 V
$\frac{1}{2}\text{Cl}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	Cl^-	1.40 V
$\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^-$	$\rightleftharpoons$	$\text{Mn}^{2+} + 4\text{H}_2\text{O}$	1.51 V
$\frac{1}{2}\text{F}_2(\text{g}) + \text{e}^-$	$\rightleftharpoons$	F^-	2.89 V

PERIODIC TABLE OF THE ELEMENTS

PERIODIC TABLE OF THE ELEMENTS																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											
1 H 1.008 Hydrogen		KEY																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
		Atomic Number		Symbol of element		Atomic Weight		Name of element		Atomic Number		Symbol of element		Atomic Weight		Name of element																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											
		79		Au		197.0		Gold		79		Au		197.0		Gold																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											
3 Li 6.941 Lithium		4 Be 9.012 Beryllium		5 B 10.81 Boron		6 C 12.01 Carbon		7 N 14.01 Nitrogen		8 O 16.00 Oxygen		9 F 19.00 Fluorine		10 Ne 20.18 Neon																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																													
11 Na 22.99 Sodium		12 Mg 24.31 Magnesium		13 Al 26.98 Aluminium		14 Si 28.09 Silicon		15 P 30.97 Phosphorus		16 S 32.07 Sulfur		17 Cl 35.45 Chlorine		18 Ar 39.95 Argon																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																													
19 K 39.10 Potassium		20 Ca 40.08 Calcium		21 Sc 44.96 Scandium		22 Ti 47.87 Titanium		23 V 50.94 Vanadium		24 Cr 52.00 Chromium		25 Mn 54.94 Manganese		26 Fe 55.85 Iron		27 Co 58.93 Cobalt		28 Ni 58.69 Nickel		29 Cu 63.55 Copper		30 Zn 65.39 Zinc		31 Ga 69.72 Gallium		32 Ge 72.61 Germanium		33 As 74.92 Arsenic		34 Se 78.96 Selenium		35 Br 79.90 Bromine		36 Kr 83.80 Krypton		37 Rb 85.47 Rubidium		38 Sr 87.62 Strontium		39 Y 88.91 Yttrium		40 Zr 91.22 Zirconium		41 Nb 92.91 Niobium		42 Mo 95.94 Molybdenum		43 Tc [98.91] Technetium		44 Ru 101.1 Ruthenium		45 Rh 102.9 Rhodium		46 Pd 106.4 Palladium		47 Ag 107.9 Silver		48 Cd 112.4 Cadmium		49 In 114.8 Indium		50 Sn 118.7 Tin		51 Sb 121.8 Antimony		52 Te 127.6 Tellurium		53 I 126.9 Iodine		54 Xe 131.3 Xenon		55 Cs 132.9 Caesium		56 Ba 137.3 Barium		57–71 Lanthanides Actinides		72 Hf 178.5 Hafnium		73 Ta 180.9 Tantalum		74 W 183.8 Tungsten		75 Re 186.2 Rhenium		76 Os 190.2 Osmium		77 Ir 192.2 Iridium		78 Pt 195.1 Platinum		79 Au 197.0 Gold		80 Hg 200.6 Mercury		81 Tl 204.4 Thallium		82 Pb 207.2 Lead		83 Bi 209.0 Bismuth		84 Po [210.0] Polonium		85 At [210.0] Astatine		86 Rn [222.0] Radon		87 Fr [223.0] Francium		88 Ra [226.0] Radium		89–103 Lanthanides Actinides		104 Rf [261.1] Rutherfordium		105 Db [262.1] Dubnium		106 Sg [263.1] Seaborgium		107 Bh [264.1] Bohrium		108 Hs [265.1] Hassium		109 Mt [268] Meitnerium		110 Uun — Ununilium		111 Uuu — Unununium		112 Uub — Ununbium		113 — — Ununtrium		114 Uuq — Ununquadium		115 — — Ununpentium		116 Uuh — Ununhexium		117 — — Ununseptium		118 Uuo — Ununoctium		119 — — Ununennium		120 — — Unbinilium		121 — — Unbinilium		122 — — Unbinilium		123 — — Unbinilium		124 — — Unbinilium		125 — — Unbinilium		126 — — Unbinilium		127 — — Unbinilium		128 — — Unbinilium		129 — — Unbinilium		130 — — Unbinilium		131 — — Unbinilium		132 — — Unbinilium		133 — — Unbinilium		134 — — Unbinilium		135 — — Unbinilium		136 — — Unbinilium		137 — — Unbinilium		138 — — Unbinilium		139 — — Unbinilium		140 — — Unbinilium		141 — — Unbinilium		142 — — Unbinilium		143 — — Unbinilium		144 — — Unbinilium		145 — — Unbinilium		146 — — Unbinilium		147 — — Unbinilium		148 — — Unbinilium		149 — — Unbinilium		150 — — Unbinilium		151 — — Unbinilium		152 — — Unbinilium		153 — — Unbinilium		154 — — Unbinilium		155 — — Unbinilium		156 — — Unbinilium		157 — — Unbinilium		158 — — Unbinilium		159 — — Unbinilium		160 — — Unbinilium		161 — — Unbinilium		162 — — Unbinilium		163 — — Unbinilium		164 — — Unbinilium		165 — — Unbinilium		166 — — Unbinilium		167 — — Unbinilium		168 — — Unbinilium		169 — — Unbinilium		170 — — Unbinilium		171 — — Unbinilium		172 — — Unbinilium		173 — — Unbinilium		174 — — Unbinilium		175 — — Unbinilium		176 — — Unbinilium		177 — — Unbinilium		178 — — Unbinilium		179 — — Unbinilium		180 — — Unbinilium		181 — — Unbinilium		182 — — Unbinilium		183 — — Unbinilium		184 — — Unbinilium		185 — — Unbinilium		186 — — Unbinilium		187 — — Unbinilium		188 — — Unbinilium		189 — — Unbinilium		190 — — Unbinilium		191 — — Unbinilium		192 — — Unbinilium		193 — — Unbinilium		194 — — Unbinilium		195 — — Unbinilium		196 — — Unbinilium		197 — — Unbinilium		198 — — Unbinilium		199 — — Unbinilium		200 — — Unbinilium		201 — — Unbinilium		202 — — Unbinilium		203 — — Unbinilium		204 — — Unbinilium		205 — — Unbinilium		206 — — Unbinilium		207 — — Unbinilium		208 — — Unbinilium		209 — — Unbinilium		210 — — Unbinilium		211 — — Unbinilium		212 — — Unbinilium		213 — — Unbinilium		214 — — Unbinilium		215 — — Unbinilium		216 — — Unbinilium		217 — — Unbinilium		218 — — Unbinilium		219 — — Unbinilium		220 — — Unbinilium		221 — — Unbinilium		222 — — Unbinilium		223 — — Unbinilium		224 — — Unbinilium		225 — — Unbinilium		226 — — Unbinilium		227 — — Unbinilium		228 — — Unbinilium		229 — — Unbinilium		230 — — Unbinilium		231 — — Unbinilium		232 — — Unbinilium		233 — — Unbinilium		234 — — Unbinilium		235 — — Unbinilium		236 — — Unbinilium		237 — — Unbinilium		238 — — Unbinilium		239 — — Unbinilium		240 — — Unbinilium		241 — — Unbinilium		242 — — Unbinilium		243 — — Unbinilium		244 — — Unbinilium		245 — — Unbinilium		246 — — Unbinilium		247 — — Unbinilium		248 — — Unbinilium		249 — — Unbinilium		250 — — Unbinilium		251 — — Unbinilium		252 — — Unbinilium		253 — — Unbinilium		254 — — Unbinilium		255 — — Unbinilium		256 — — Unbinilium		257 — — Unbinilium		258 — — Unbinilium		259 — — Unbinilium		260 — — Unbinilium		261 — — Unbinilium		262 — — Unbinilium		263 — — Unbinilium		264 — — Unbinilium		265 — — Unbinilium		266 — — Unbinilium		267 — — Unbinilium		268 — — Unbinilium		269 — — Unbinilium		270 — — Unbinilium		271 — — Unbinilium		272 — — Unbinilium		273 — — Unbinilium		274 — — Unbinilium		275 — — Unbinilium		276 — — Unbinilium		277 — — Unbinilium		278 — — Unbinilium		279 — — Unbinilium		280 — — Unbinilium		281 — — Unbinilium		282 — — Unbinilium		283 — — Unbinilium		284 — — Unbinilium		285 — — Unbinilium		286 — — Unbinilium		287 — — Unbinilium		288 — — Unbinilium		289 — — Unbinilium		290 — — Unbinilium		291 — — Unbinilium		292 — — Unbinilium		293 — — Unbinilium		294 — — Unbinilium		295 — — Unbinilium		296 — — Unbinilium		297 — — Unbinilium		298 — — Unbinilium		299 — — Unbinilium		300 — — Unbinilium		301 — — Unbinilium		302 — — Unbinilium		303 — — Unbinilium		304 — — Unbinilium		305 — — Unbinilium		306 — — Unbinilium		307 — — Unbinilium		308 — — Unbinilium		309 — — Unbinilium		310 — — Unbinilium		311 — — Unbinilium		312 — — Unbinilium		313 — — Unbinilium		314 — — Unbinilium		315 — — Unbinilium		316 — — Unbinilium		317 — — Unbinilium		318 — — Unbinilium		319 — — Unbinilium		320 — — Unbinilium		321 — — Unbinilium		322 — — Unbinilium		323 — — Unbinilium		324 — — Unbinilium		325 — — Unbinilium		326 — — Unbinilium		327 — — Unbinilium		328 — — Unbinilium		329 — — Unbinilium		330 — — Unbinilium		331 — — Unbinilium		332 — — Unbinilium		333 — — Unbinilium		334 — — Unbinilium		335 — — Unbinilium		336 — — Unbinilium		337 — — Unbinilium		338 — — Unbinilium		339 — — Unbinilium		340 — — Unbinilium		341 — — Unbinilium		342 — — Unbinilium		343 — — Unbinilium		344 — — Unbinilium		345 — — Unbinilium		346 — — Unbinilium		347 — — Unbinilium		348 — — Unbinilium		349 — — Unbinilium		350 — — Unbinilium		351 — — Unbinilium		352 — — Unbinilium		353 — — Unbinilium		354 — — Unbinilium		355 — — Unbinilium		356 — — Unbinilium		357 — — Unbinilium		358 — — Unbinilium		359 — — Unbinilium		360 — — Unbinilium		361 — — Unbinilium		362 — — Unbinilium		363 — — Unbinilium		364 — — Unbinilium		365 — — Unbinilium		366 — — Unbinilium		367 — — Unbinilium		368 — — Unbinilium		369 — — Unbinilium		370 — — Unbinilium		371 — — Unbinilium		372 — — Unbinilium		373 — — Unbinilium		374 — — Unbinilium		375 — — Unbinilium		376 — — Unbinilium		377 — — Unbinilium		378 — — Unbinilium		379 — — Unbinilium		380 — — Unbinilium		381 — — Unbinilium		382 — — Unbinilium		383 — — Unbinilium		384 — — Unbinilium		385 — — Unbinilium		386 — — Unbinilium		387 — — Unbinilium		388 — — Unbinilium		389 — — Unbinilium		390 — — Unbinilium		391 — — Unbinilium		392 — — Unbinilium		393 — — Unbinilium		394 — — Unbinilium		395 — — Unbinilium		396 — — Unbinilium		397 — — Unbinilium		398 — — Unbinilium		399 — — Unbinilium		400 — — Unbinilium		401 — — Unbinilium		402 — — Unbinilium		403 — — Unbinilium		404 — — Unbinilium		405 — — Unbinilium		406 — — Unbinilium		407 — — Unbinilium		408 — — Unbinilium		409 — — Unbinilium		410 — — Unbinilium		411 — — Unbinilium		412 — — Unbinilium		413 — — Unbinilium		414 — — Unbinilium		415 — — Unbinilium		416 — — Unbinilium		417 — — Unbinilium		418 — — Unbinilium		419 — — Unbinilium		420 — — Unbinilium		421 — — Unbinilium		422 — — Unbinilium		423 — — Unbinilium		424 — — Unbinilium		425 — — Unbinilium		426 — — Unbinilium		427 — — Unbinilium		428 — — Unbinilium		429 — — Unbinilium		430 — — Unbinilium		431 — — Unbinilium		432 — — Unbinilium		433 — — Unbinilium		434 — — Unbinilium		435 — — Unbinilium		436 — — Unbinilium		437 — — Unbinilium		438 — — Unbinilium		439 — — Unbinilium		440 — — Unbinilium		441 — — Unbinilium		442 — — Unbinilium		443 — — Unbinilium		444 — — Unbinilium		445 — — Unbinilium		446 — — Unbinilium		447 — — Unbinilium		448 — — Unbinilium		449 — — Unbinilium		450 — — Unbinilium		451 — — Unbinilium		452 — — Unbinilium		453 — — Unbinilium		454 — — Unbinilium		455 — — Unbinilium		456 — — Unbinilium		457 — — Unbinilium		458 — — Unbinilium		459 — — Unbinilium		460 — — Unbinilium		461 — — Unbinilium		462 — — Unbinilium		463 — — Unbinilium		464 — — Unbinilium		465 — — Unbinilium		466 — — Unbinilium		467 — — Unbinilium		468 — — Unbinilium		469 — — Unbinilium		470 — — Unbinilium		471 — — Unbinilium		472 — — Unbinilium		473 — — Unbinilium		474 — — Unbinilium		475 — — Unbinilium		476 — — Unbinilium		477 — — Unbinilium		478 — — Unbinilium		479 — — Unbinilium		480 — — Unbinilium		481 — — Unbinilium		482 — — Unbinilium		483 — — Unbinilium		484 — — Unbinilium		485 — — Unbinilium		486 — — Unbinilium		487 — — Unbinilium		488 — — Unbinilium		489 — — Unbinilium		490 — — Unbinilium		491 — — Unbinilium		492 — — Unbinilium		493 — — Unbinilium		494 — — Unbinilium		495 — — Unbinilium		496 — — Unbinilium		497 — — Unbinilium		498 — — Unbinilium		499 — — Unbinilium		500 — — Unbinilium		501 — — Unbinilium		502 — — Unbinilium		503 — — Unbinilium		504 — — Unbinilium		505 — — Unbinilium		506 — — Unbinilium		507 — — Unbinilium		508 — — Unbinilium		509 — — Unbinilium		510 — — Unbinilium		511 — — Unbinilium		512 — — Unbinilium		513 — — Unbinilium		514 — — Unbinilium		515 — — Unbinilium		516 — 	

Lanthanides

57 La 138.9 Lanthanum	58 Ce 140.1 Cerium	59 Pr 140.9 Praseodymium	60 Nd 144.2 Neodymium	61 Pm [146.9] Promethium	62 Sm 150.4 Samarium	63 Eu 152.0 Europium	64 Gd 157.3 Gadolinium	65 Tb 158.9 Terbium	66 Dy 162.5 Dysprosium	67 Ho 164.9 Holmium	68 Er 167.3 Erbium	69 Tm 168.9 Thulium	70 Yb 173.0 Ytterbium	71 Lu 175.0 Lutetium
--------------------------------	-----------------------------	-----------------------------------	--------------------------------	-----------------------------------	-------------------------------	-------------------------------	---------------------------------	------------------------------	---------------------------------	------------------------------	-----------------------------	------------------------------	--------------------------------	-------------------------------

Actinides

89 Ac [227.0] Actinium	90 Th 232.0 Thorium	91 Pa 231.0 Protactinium	92 U 238.0 Uranium	93 Np [237.0] Neptunium	94 Pu [239.1] Plutonium	95 Am [241.1] Americium	96 Cm [244.1] Curium	97 Bk [249.1] Berkelium	98 Cf [252.1] Californium	99 Es [252.1] Einsteinium	100 Fm [257.1] Fermium	101 Md [258.1] Mendelevium	102 No [259.1] Nobelium	103 Lr [262.1] Lawrencium
---------------------------------	------------------------------	-----------------------------------	-----------------------------	----------------------------------	----------------------------------	----------------------------------	-------------------------------	----------------------------------	------------------------------------	------------------------------------	---------------------------------	-------------------------------------	----------------------------------	------------------------------------

Where the atomic weight is not known, the relative atomic mass of the most common radioactive isotope is shown in brackets.
The atomic weights of Np and Tc are given for the isotopes ²³⁷Np and ⁹⁹Tc.

**2003 HSC Notes from
the Marking Centre
Chemistry**

© 2004 Copyright Board of Studies NSW for and on behalf of the Crown in right of the State of New South Wales.

This document contains Material prepared by the Board of Studies NSW for and on behalf of the State of New South Wales. The Material is protected by Crown copyright.

All rights reserved. No part of the Material may be reproduced in Australia or in any other country by any process, electronic or otherwise, in any material form or transmitted to any other person or stored electronically in any form without the prior written permission of the Board of Studies NSW, except as permitted by the *Copyright Act 1968*. School candidates in NSW and teachers in schools in NSW may copy reasonable portions of the Material for the purposes of bona fide research or study.

When you access the Material you agree:

- to use the Material for information purposes only
- to reproduce a single copy for personal bona fide study use only and not to reproduce any major extract or the entire Material without the prior permission of the Board of Studies NSW
- to acknowledge that the Material is provided by the Board of Studies NSW
- not to make any charge for providing the Material or any part of the Material to another person or in any way make commercial use of the Material without the prior written consent of the Board of Studies NSW and payment of the appropriate copyright fee
- to include this copyright notice in any copy made
- not to modify the Material or any part of the Material without the express prior written permission of the Board of Studies NSW.

The Material may contain third party copyright materials such as photos, diagrams, quotations, cartoons and artworks. These materials are protected by Australian and international copyright laws and may not be reproduced or transmitted in any format without the copyright owner's specific permission. Unauthorised reproduction, transmission or commercial use of such copyright materials may result in prosecution.

The Board of Studies has made all reasonable attempts to locate owners of third party copyright material and invites anyone from whom permission has not been sought to contact the Copyright Officer, ph (02) 9367 8289, fax (02) 9279 1482.

Published by Board of Studies NSW
GPO Box 5300
Sydney 2002
Australia

Tel: (02) 9367 8111

Fax: (02) 9367 8484

Internet: www.boardofstudies.nsw.edu.au

ISBN 1 7409 9988 6

2004034

Contents

Section I – Core 6

Section II – Options 9

2003 HSC NOTES FROM THE MARKING CENTRE

CHEMISTRY

Introduction

This document has been produced for the teachers and candidates of the Stage 6 course in Chemistry. It provides comments with regard to responses to the 2003 Higher School Certificate Examination, indicating the quality of candidate responses and highlighting the relative strengths and weaknesses of the candidature in each section and each question.

It is essential for this document to be read in conjunction with the relevant syllabus, the 2003 Higher School Certificate Examination, the Marking Guidelines and other support documents which have been developed by the Board of Studies to assist in the teaching and learning of Chemistry.

General Comments

In 2003, 9328 candidates attempted the chemistry examination.

This year candidates seemed to have a greater awareness of the importance of the key words in the examination. Spelling, grammar and scientific expression were very poor from some candidates, with handwriting being more illegible than in the past.

Candidates need to read the question thoroughly and respond to each part. The best responses provided evidence that the candidate had planned out an answer to fit into the space provided. Responses that used extra booklets rarely provided additional worthwhile material.

Candidates need to set out their work clearly in calculation questions. Where appropriate, it is advisable to write balanced chemical equations, including the correct states of matter, to support an answer.

The variation in the quality of responses seems to suggest that candidates who have actively planned and performed a first-hand investigation are more likely to have retained knowledge and understanding of the experience than those who have rote learnt an expected answer. In addition, the development of the Chemistry skills will be more evident and rewarded with marks.

Teachers and candidates should be aware that examiners may write questions that address the syllabus outcomes in a manner that requires candidates to respond by integrating their knowledge, understanding and skills developed through studying the course. This reflects the fact that the knowledge, understanding and skills developed through the study of discrete sections should accumulate to a more comprehensive understanding than may be described in each section separately. This aspect needs to be more fully appreciated by all systems and candidates.

Section I – Core**Part A – Multiple choice**

Question	Correct Response
1	B
2	D
3	B
4	B
5	D
6	C
7	C
8	A

Question	Correct Response
9	A
10	A
11	B
12	D
13	D
14	C
15	A

Part B**General Comments**

Overall, the candidates' responses indicated that the majority had a grasp of chemical concepts, appropriate for HSC candidates.

Specific Comments**Question 16**

- (a) Many candidates did not provide a valid procedure for this first hand investigation.
- (b) The equation was generally well done, although many candidates ignored, or used incorrect, states of matter.

Question 17

- (a) The identification was well done.
- (b) The best responses provided the features of a process and included a thorough knowledge of the three step radical addition polymerisation or the Ziegler-Natta catalysis or of the conditions needed in gas-phase polymerisation.
- (c) The better responses gave a reason for the distribution of molecular weights.

Question 18

Better responses demonstrated a thorough knowledge of transuranic elements, commercial radioisotopes, nuclear reactors, accelerators and the bombarding particles required in each case. Many candidates knew one specific radioisotope but few could define a transuranic element.

Question 19

- (a) The majority of the candidates correctly identified the silver cathode; however, a significant number nominated lead with some incorrectly identifying an ion as the cathode.
- (b) Most responses demonstrated that candidates can transfer information from the Data Sheet. Poorer responses did not include balanced equations with the correct number of electrons and subscripts were omitted from the net equation.

Better responses demonstrated that the value should be positive for the half-cell reactions. It is advised that if states of matter are not shown in the equation that the aqueous state is assumed. In this question the identification of the solid state is an essential part of the chemistry required.

Question 20

Generally well answered. Most candidates used alcohol, biopolymers and ethene. Better responses introduced some potential chemicals produced from biomass and related these to the energy and industrial uses and production of these chemicals. Poor responses indicated a limited knowledge of the chemical use of biomass through the use of general statements and lack of detail or structure in their answer.

Question 21

- (a) Many responses focused on the naming of the ester and left out the water.
- (b) Best responses included a fully labelled diagram to gain full marks. Many responses indicated the lack of knowledge of the investigation with a minority of responses providing diagrams of apparatus for distillation rather than refluxing.
- (c) The better responses outlined more than one advantage of refluxing. Poorer responses suggested that candidates had written a memorised unit of information about refluxing rather than answer the question asked.

Question 22

- (a) Most candidates correctly answered the question.
- (b) A significant number of responses did not use the correct data provided on the Data Sheet. Poorer responses demonstrated little chemical knowledge of mole ratios and a lack of understanding of the need to show steps in the calculations used to achieve their answer.

Question 23

- (a) The majority of candidates answered this question correctly. The equation posed a problem for those who did not know the chemical formula for nitric acid.
- (b) Common errors included using the dilution formula, and not taking into account the mole ratio in the calculation. The better responses discarded the first titre value from their calculations.

Question 24

This question was poorly answered. Few responses demonstrated a sound understanding of this area of the syllabus, and many could not link knowledge of neutralisation to the cleaning up of a chemical spill. The best responses demonstrated a knowledge of amphoteric substances and the benefits of using a solid such as sodium hydrogen carbonate to clean up both acidic and basic spills. The majority of responses provided features of safety procedures, such as wearing safety goggles, and problems associated with run-off to the environment.

Question 25

This question was poorly done, with many responses written beyond the available space without any benefit. The better responses indicated a clear understanding of the interactions between and within molecules. Poorer responses used the term 'hydroxide ion' rather than 'hydroxy group' indicating a lack of understanding about the bonding issues involved. Many responses described the slopes and shapes of the graphs but did not relate these to reasons for the trends within and between the graphs.

Question 26

This question was poorly answered. Of serious concern was the proportion of candidates who learnt a definition given in one of the common texts which was not chemically correct. It is advisable to reinforce the need to access a 'wide variety of secondary resources' rather than rely on a single source of information. The best responses provided characteristics of eutrophication, stated two or more of the tests for BOD, DO, nitrate and phosphate, then made a judgement about the usefulness of the results gained.

Question 27

- (a) The better responses clearly set out and correctly calculated the required mass from the correct formula for barium sulfate. Poorer responses did not indicate that mass of the sulfate depended on mass of sulfate ions present not the barium ions
- (b) Better responses demonstrated knowledge of the relationship between repetition and reliability and made judgements based on their explanation

Question 28

- (a) The best responses clearly plotted the point '0,0.00' and drew a line of best fit.
- (b) The better responses related information from the graph in (a) to answer this question concisely. Many responses confused the sampling site with the source and some tried to include their understanding of causes of cadmium pollution into their responses without analysing the data provided.

Question 29

The better responses demonstrated that an evaluation of the monitoring and management of conditions in the Haber process would require a description of the process, an understanding of the chemistry behind the process and explanations of how the conditions were critical to the process.

Section II – Options

Specific Comments

Question 30 – Industrial Chemistry

- (a) (i) Most candidates correctly identified one use.
- (ii) Many candidates scored well on this part. The better responses gave evidence of up-to-date knowledge of current industrial processes and included a diagram. Some responses mixed the different ways in which the Frasch process is undertaken and created confusion over the use of terms ‘compressed air’, ‘condensed air’ and ‘superheated steam’.
- (b) (i) Most candidates gained a mark in this part. Better responses included precautions specific to the investigation in the question.
- (ii) From the responses it was evident that most candidates had completed the investigation but some confused it with the commercial production method. Many responses did not include a full description of how the hydroxide ion is identified. Many responses included a statement about the test that was used but did not clearly show the result of that test, or they gave a result but no test was identified.
- (c) This question was poorly answered. While many candidates provided a link between soaps and the development of detergents, very few candidates could give a clear analysis involving the structures of soaps and detergents as well as the cleaning action of soaps and detergents.
- (d) (i) Most candidates successfully identified temperature or change in temperature as the factor.
- (ii) A majority of candidates were able to describe two ways to increase yield. Many candidates elaborated and described, mainly using Le Chatelier’s Principle, why these would affect yield.
- (iii) Most responses included the equilibrium expression. Common errors included having the expression inverted and missing the ‘powers’. Amongst the better responses, candidates were able to attempt the calculation of equilibrium concentrations and correctly insert in the algorithm and calculate a value for K. Very few candidates could arrive at the correct final K value.
- (e) The better responses provided appropriate detail, balanced equations and included an appropriate judgement.
- Poorer responses identified an environmental issue or outlined how one environmental issue was associated with an industrial method of preparation of either an acid or a base. In some cases, their knowledge was good for one method but weak for another.
- Many responses lacked any detail as to why certain products actually caused problems.

Question 31 – Shipwrecks, Corrosion and Conservation

- (a) (i) This was well answered with the majority of candidates being able to identify one passivating metal.
- (ii) Many responses showed a basic understanding of active and passivating metals. They

generally understood the concept of non-protective layers but did not always attribute it to the formation of an oxide layer. There was poor use of technical language, e.g. porous/non-porous; pervious/ impervious. Aluminium and iron were the common examples cited. A significant number of candidates did not know that both form M(III) oxides. The required responses included a balanced equation or half-equation. There was confusion between 'active' relating to the activity series of metals and 'active' meaning 'not passivating'.

- (b) (i) The better responses indicated precautions particular to their investigation, eg copper sulfate was placed in a waste bottle. Poorer responses focused on the possible dangers without considering appropriate strategies to minimise them.
- (ii) This part of the question assessed practical work undertaken in the course, and therefore responses would include the particular electrolyte and electrodes used. It was also expected that they indicate the factor they were testing and how this could be measured or observed while controlling other factors. Many candidates indicated that 'electrodes were placed in an electrolyte' without being specific. Poorer responses did not adequately distinguish between factors affecting the rate of electrolysis and factors affecting the products produced. The use of a labelled diagram wherever possible is to be recommended.
- (c) The better responses included 'old theories' and 'more recent theories' and included a number of significant factors such as:
- 'old theories' gave reasons for very low oxygen concentration, hence there was expected to be very little corrosion.
 - 'more recent theories' account for the presence of sulfate-producing bacteria which still allows for the oxidation of iron to occur since the sulfate is being reduced. Half equations showing this process were included.

While many responses mentioned anaerobic bacteria some thought they produced oxygen which contributed to corrosion: therefore no change to the theory.

- (d) (i) Most responses identified iron as the correct metal.
- (ii) Most candidates found plastic difficult to deal with in the context of oxidation/reduction, as did they with the idea of 'limiting'. Some candidates reversed Cu/Al as more active/less active. Many candidates seemed confused by aluminium as active and /or passivating with respect to the corrosion of the hooks. In addition, many did not write about the corrosion of the hooks, only the containers. Only a small percentage of candidates used chemical equations in their responses and overall the use of chemical terminology, eg reductant, cathodic, etc, was poor.
- (e) The best responses included an appropriate judgement about the use of electrolysis in cleaning and restoring artefacts, and demonstrated a thorough knowledge of it. Most responses did not distinguish between cleaning and restoring artefacts.

Question 32 – Biochemistry of Movement

- (a) Most candidates answered both parts of the question correctly.
- (b) (i) The better responses referred to their use of diagrams or models or these molecules.

- (ii) Many candidates did not attempt this part at all whilst some responses included discussion of the polarity of the -OH bonds and hydrogen bonding.
- (c) Most candidates answered this part poorly. The better responses demonstrated understanding of the sliding filament theory and the structure of proteins.
- (d) (i) Most candidates answered this question correctly.
 - (ii) The better responses included word equations and demonstrated the difference between ADP and ATP.
 - (iii) Most responses to this part were poor. Candidates found it difficult to relate the role of the oxidation levels of the fats and carbohydrates to energy.
- (e) The better responses demonstrated an extensive knowledge of two or more metabolic pathways. They showed an understanding of the unique role of ATP and made a judgement about its input in metabolic pathways.

Question 33 – Chemistry of Art

- (a) (i) Generally well answered; however, a small number confused an orbital with a shell and some gave the answer ' $2n^2$ '.
 - (ii) While most responses focused on the jump in ionisation energy when an electron was removed from an inner shell, few did not relate this to the general increase in ionisation energies. Most responses described the graph but few answers related the increasing electrostatic attraction between inner shell electrons and the nucleus.
- (b) (i) and (ii) Both parts were well done. The best responses stated a precaution and related this to the investigation. It was evident that a great diversity of investigations had been carried out.
- (c) Most candidates had a good knowledge of the pigments used by Aboriginal people. Many candidates could give a property other than colour eg solubility, but few explained why this property was a feature of these pigments. Few responses explained the relationship between chemical composition and colour.
- (d) (i) Extremely well done.
 - (ii) Generally well done. The best responses discussed oxidation and demonstrated an understanding of oxidation state.
 - (iii) The best responses gave a good explanation of the chemistry of chelation. Poorer responses indicated the role of the non-bonding electron pair was not well understood.
- (e) The better responses were concise and showed evidence of planning to answer the question being asked. They included good descriptions of two or more technologies, their uses, advantages and disadvantages, then making a judgement about their usefulness in pigment identification.

Question 34 – Forensic Chemistry

- (a) (i) The majority of candidates answered this correctly.
- (ii) Some candidates tried to show a distinction between all of the classes of compounds indicated. Better responses clearly stated which classes of organic compounds were being distinguished with which test and the results of the tests.
- (b) (i) Many responses adequately identified a safety precaution. Better responses included precautions specific to the investigation in the question.
- (ii) The better responses outlined an identifying test to distinguish between reducing and non-reducing sugars, giving examples of each.
- (c) Better responses demonstrated an extensive knowledge of emission spectra analysis of a mixture. The majority of the responses displayed a sound knowledge of the use of emission spectra analysis to identify elements.
- (d) (i) Virtually all candidates were able to identify F2 as the more probable father.
- (ii) Generally well done. Better responses demonstrated a sound knowledge of DNA structure and composition.
- (iii) This part was not well answered by the majority of candidates. The better responses correctly referred to the importance of the non-coding sections of DNA as being important in the analysis of DNA. Few candidates referred to the stimulus material in their response.
- (e) Generally well done. A number of candidates displayed a sound knowledge of chromatographic principles and applications. Better responses demonstrated a high appreciation of the developments in chromatography to advance Forensic Chemistry through the evaluation of methods in comparison to simpler methods and described the principles of these methods.

Chemistry

2003 HSC Examination Mapping Grid

Question	Marks	Content	Syllabus outcomes
Section I Part A			
1	1	9.3.1	H8, H9
2	1	9.1, 9.2.5	H12
3	1	9.1, 9.2.5	H3, H11
4	1	9.4.4	H4
5	1	9.1, 9.4.5	H11
6	1	9.1, 9.2.3	H7, H9, H14
7	1	9.1, 9.3.2	H6, H14
8	1	9.1, 9.3.3	H8, H10, H12
9	1	9.4.4	H9, H12
10	1	9.1, 9.4.4	H14
11	1	9.1, 9.2.1	H9, H13
12	1	9.1, 9.4.4	H10, H12, H14
13	1	9.1, 9.2.4	H6, H8, H14
14	1	9.1, 9.3.4	H8, H11, H12
15	1	9.1, 9.3.4	H8, H10, H14

Question	Marks	Content	Syllabus outcomes
Section I Part B			
16 (a)	2	9.1, 9.2.3	H11
16 (b)	1	9.1, 9.2.3	H9, H10, H13
17 (a)	1	9.1, 9.2.3	H8, H9, H12
17 (b)	3	9.2.1	H3, H7, H8, H9, H10, H13
17 (c)	1	9.2.1	H9, H12
18	4	9.2.5	H3, H5, H8
19 (a)	1	9.1, 9.2.4	H11, H13, H14
19 (b)	2	9.1, 9.2.4	H7, H8, H10, H12
20	5	9.2.2	H1, H4, H5, H6, H8, H9, H13
21 (a)	1	9.2.2, 9.2.3, 9.3.5	H8, H9
21 (b)	2	9.1, 9.3.5	H11, H13
21 (c)	2	9.1, 9.3.5	H11, H12, H13
22 (a)	1	9.2.1, 9.2.3	H9, H10, H13
22 (b)	2	9.3.2	H7, H8, H9, H10, H12, H13
23 (a)	1	9.2.1	H10, H13
23 (b)	3	9.3.4	H8, H10, H11, H12, H13
24	4	9.3.3, 9.3.4	H4, H7, H8, H11
25	4	9.1, 9.3.5	H6, H12, H14
26	4	9.1, 9.4.5	H3, H5, H7, H8, H9, H11, H14
27 (a)	2	9.1, 9.4.3	H10, H12, H13
27 (b)	3	9.1, 9.4.3	H10, H12, H14
28 (a)	2	9.1, 9.4.3	H12, H13
28 (b)	2	9.1, 9.4.3	H4, H12, H14
29	7	9.3.2, 9.4.1, 9.4.2	H1, H2, H3, H4, H7, H8, H10

Question	Marks	Content	Syllabus outcomes
Section II			
Question 30 — Industrial Chemistry			
30 (a) (i)	1	9.5.3	H6
30 (a) (ii)	3	9.5.3	H8
30 (b) (i)	1	9.1	H8, H12, H13
30 (b) (ii)	3	9.5.4	H8, H12, H13
30 (c)	5	9.5.5	H1, H4, H9, H13
30 (d) (i)	1	9.5.2	H8
30 (d) (ii)	2	9.5.2	H8
30 (d) (iii)	3	9.5.2	H8, H10, H12
30 (e)	6	9.5.3, 9.5.4, 9.5.6	H4, H8, H13
Section II			
Question 31 — Shipwrecks, Corrosion and Conservation			
31 (a) (i)	1	9.6.2	H8
31 (a) (ii)	3	9.6.2	H6, H8, H13
31 (b) (i)	1	9.1	H8, H12, H13
31 (b) (ii)	3	9.6.3	H8, H12, H13
31 (c)	5	9.6.5, 9.6.6	H1, H2, H8, H13
31 (d) (i)	1	9.6.2	H8
31 (d) (ii)	5	9.6.4	H4, H8, H13, H14
31 (e)	6	9.6.1, 9.6.2, 9.6.3, 9.6.7	H3, H7, H8, H13
Section II			
Question 32 — The Biochemistry of Movement			
32 (a) (i)	1	9.7.2	H4, H9
32 (a) (ii)	3	9.7.2	H6, H9, H13
32 (b) (i)	2	9.7.3	H2, H9, H12, H13
32 (b) (ii)	2	9.7.3	H9
32 (c)	5	9.7.4, 9.7.5	H1, H7, H8, H9, H13
32 (d) (i)	1	9.7.7, 9.7.8, 9.7.9	H7, H9

Question	Marks	Content	Syllabus outcomes
32 (d) (ii)	2	9.7.1, 9.7.7	H7, H9, H10
32 (d) (iii)	3	9.7.3, 9.7.6	H2, H8, H9, H10, H12, H13
32 (e)	6	9.7.1, 9.7.3, 9.7.5, 9.7.6, 9.7.7, 9.7.8, 9.7.9, 9.7.10	H1, H7, H8, H9, H10
Section II			
Question 33 — The Chemistry of Art			
33 (a) (i)	1	9.8.3	H1, H6
33 (a) (ii)	3	9.8.3	H7
33 (b) (i)	2	9.1	H8, H12, H13
33 (b) (ii)	2	9.8.2	H8, H12, H13
33 (c)	5	9.8.1, 9.8.4, 9.8.5	H1, H2, H4
33 (d) (i)	1	9.8.4	H6
33 (d) (ii)	2	9.8.4	H6, H7, H8
33 (d) (iii)	3	9.8.5	H3, H4, H6, H8, H13
33 (e)	6	9.8.1, 9.8.2, 9.8.3	H1, H4, H6, H7, H13
Section II			
Question 34 — Forensic Chemistry			
34 (a) (i)	1	9.9.2	H9, H10
34 (a) (ii)	3	9.9.1	H8, H9
34 (b) (i)	1	9.1	H12, H13
34 (b) (ii)	3	9.9.2	H12, H13
34 (c)	5	9.9.6	H3, H4, H6, H7, H13
34 (d) (i)	1	9.9.4	H4, H12, H14
34 (d) (ii)	2	9.9.4	H2, H9, H10, H13
34 (d) (iii)	3	9.9.1, 9.9.4	H1, H3, H4, H5, H13
34 (e)	6	9.9.3, 9.9.5	H1, H3, H4, H8, H9, H13

2003 HSC Chemistry Marking Guidelines

Question 16 (a)

Outcomes assessed: H11

MARKING GUIDELINES

Criteria	Marks
• Correctly outlines the procedure for monitoring mass changes	2
• Partially outlines a procedure for monitoring mass changes	1

Question 16 (b)

Outcomes assessed: H9, H10, H13

MARKING GUIDELINES

Criteria	Marks
• Correctly writes the balanced equation	1

Question 17 (a)

Outcomes assessed: H8, H9, H12

MARKING GUIDELINES

Criteria	Marks
• Correctly identifies ethanol by name or chemical formula	1

Question 17 (b)

Outcomes assessed: H3, H7, H8, H9, H10, H13

MARKING GUIDELINES

Criteria	Marks
Provides a thorough description of the three key steps in free radical addition polymerisation or the steps in Ziegler/Natta polymerisation	3
- Provides a sound description of one step in either free radical addition polymerisation or Ziegler/Natta polymerisation OR - Provides a description of at least two conditions and reagents involved in either free radical addition polymerisation or Ziegler/Natta polymerisation	2
- Identifies an aspect of either free radical addition polymerisation or Ziegler/Natta polymerisation OR - High temperature/high pressure conditions OR - Identifies the three steps in free radical polymerisation OR - Simplified model of addition polymerisation	1

Question 17 (c)

Outcomes assessed: H9, H12

MARKING GUIDELINES

Criteria	Marks
Gives a reason for the distribution	1

Question 18

Outcomes assessed: H3, H5, H8

MARKING GUIDELINES

Criteria	Marks
Demonstrates a thorough knowledge of methods of production of both transuranic elements and commercial radioisotopes	4
Demonstrates a sound knowledge of methods of production of both transuranic elements and commercial radioisotopes	2–3
- Identifies the location of the transuranic elements in the Periodic Table OR - Identifies a transuranic element and/or commercial radioisotope and its use OR - Identifies a method or piece of equipment used to produce transuranic elements and/or commercial radioisotopes	1

Question 19 (a)

Outcomes assessed: H11, H13, H14

MARKING GUIDELINES

Criteria	Marks
Identifies cathode	1

Question 19 (b)

Outcomes assessed: H7, H8, H10, H12

MARKING GUIDELINES

Criteria	Marks
- Writes correct net redox equation including correct states of matter - Calculates correct potential	2
- Writes correct net redox equation including correct states of matter OR - Calculates correct potential	1

Question 20

Outcomes assessed: H1, H4, H5, H6, H8, H9, H13

MARKING GUIDELINES

Criteria	Marks
- Demonstrates a thorough knowledge of how biomass can be used to provide energy and chemicals - Provides an assessment of problem and/or a benefit	4–5
- Demonstrates a sound knowledge of why alternative compounds are needed OR - Identifies compounds derived from biomass - Provides an explanation of problems or benefits associated with the use of biomass	2–3
- Identifies a compound derived from biomass OR - Identifies cellulose as the main component of biomass OR - Identifies one reason for needing an alternative source of chemicals and/or energy OR - Identifies one advantage of using biomass OR - Identifies one disadvantage of using biomass	1

Question 21 (a)

Outcomes assessed: H8, H9

MARKING GUIDELINES

Criteria	Marks
Names or writes correct formulas (not molecular) for BOTH ester AND water	1

Question 21 (b)

Outcomes assessed: H11, H13

MARKING GUIDELINES

Criteria	Marks
Draws correctly labelled diagram of apparatus for reflux (essentials identified)	2
- Draws a partially correct, labelled diagram of apparatus of reflux (heat source, condenser, water flow) OR - Appropriate reflux arrangement implied with 3 of 6 essentials OR - Equipment drawn correctly	1

Question 21 (c)

Outcomes assessed: H11, H12, H13

MARKING GUIDELINES

Criteria	Marks
Outlines two advantages of reflux	2
Outlines one advantage of using reflux	1

Question 22 (a)

Outcomes assessed: H9, H10, H13

MARKING GUIDELINES

Criteria	Marks
Writes the correct balanced chemical equation	1

Question 22 (b)

Outcomes assessed: H7, H8, H9, H10, H12, H13

MARKING GUIDELINES

Criteria	Marks
Correctly calculates volume of carbon dioxide (working to be shown)	2
Shows one correct step in the calculation	1

Question 23 (a)

Outcomes assessed: H10, H13

MARKING GUIDELINES

Criteria	Marks
Provides the correct balanced equation	1

Question 23 (b)

Outcomes assessed: H8, H10, H11, H12, H13

MARKING GUIDELINES

Criteria	Marks
- Accurately calculates the concentration of nitric acid as 0.33 mol L^{-1} - Rejects the first titre in calculating the average volume	3
- Calculates the concentration of nitric acid as 0.32 mol L^{-1} (Student did not eliminate the first titre in calculating the average volume) OR - Calculates the concentration of nitric acid as 0.3 mol L^{-1} OR - Makes one error in calculating the concentration of nitric acid OR - Calculates the moles of nitric acid correctly	2
- Writes a correct, relevant mathematical formula (eg moles = molarity $\times$ volume) OR - Calculates an average titre (including or excluding first titre)	1

Question 24

Outcomes assessed: H4, H7, H8, H11

MARKING GUIDELINES

Criteria	Marks
Demonstrates a thorough knowledge of the factors and issues that must be considered when using neutralisation reactions to safely minimise damage in chemical spills	3–4
States some factors that must be considered when using neutralisation reactions	1–2

Question 25

Outcomes assessed: H6, H12, H14

MARKING GUIDELINES

Criteria	Marks
Provides a thorough explanation of the trends in boiling points BETWEEN AND WITHIN the three series of compounds	4
- Provides thorough explanation of the trends in boiling points BETWEEN each series OR - Provides sound explanation of the trends in boiling points BETWEEN the three series of compounds and identifies an aspect of the trends in boiling points WITHIN each series	2–3
- Identifies the relationship between boiling points and intermolecular forces OR - Identifies that boiling points increase as molecular weight increases OR - Identifies the following trend in boiling points: Alkanoic acids > Alkanols > Alkanes OR - Identifies hydrogen bonding as a strong intermolecular force	1

Question 26

Outcomes assessed: H3, H5, H7, H8, H9, H11, H14

MARKING GUIDELINES

Criteria	Marks
- Provides a thorough description of the process of eutrophication - States two or more tests used to monitor eutrophication - Provides an assessment OR - Applications of tests OR - Explains one or both tests in detail	4
- Provides a thorough description of the process of eutrophication AND - States two tests used to monitor eutrophication OR - Explains one test in detail OR - Basic description AND states two tests and an assessment - Explains why one named test is useful for monitoring eutrophication	3
- Provides a thorough description of the process of eutrophication OR - States two tests used to monitor eutrophication OR - Explains why one named test is useful for monitoring eutrophication	2
- Provides a basic description of the process of eutrophication OR - States one appropriate test used to monitor eutrophication	1

Question 27 (a)

Outcomes assessed: H10, H12, H13

MARKING GUIDELINES

Criteria	Marks
Correctly calculates the percentage of sulfate	2
Shows one correct step in the calculation	1

Question 27 (b)

Outcomes assessed: H10, H12, H14

MARKING GUIDELINES

Criteria	Marks
- Explains clearly how reliability depends on repetition of the procedure OR - Refers to dependence of reliability on repetition and discusses appropriate valid procedures - Provides an evaluation	2–3
- States dependence of reliability of the procedure on repetition OR - Discusses an appropriate valid procedure	1

Question 28 (a)

Outcomes assessed: H12, H13

MARKING GUIDELINES

Criteria	Marks
- Plots data accurately - Draws a line of best fit	2
- Plots data accurately OR - Draws line of best fit	1

Question 28 (b)

Outcomes assessed: H4, H12, H14

MARKING GUIDELINES

Criteria	Marks
- Identifies the correct source of cadmium pollution - Justifies the conclusion with reference to the data table	2
Identifies the correct source of cadmium pollution	1

Question 29

Outcomes assessed: H1, H2, H3, H4, H7, H8, H10

MARKING GUIDELINES

Criteria	Marks
• Demonstrates an extensive knowledge of the Haber process including factors that are monitored and managed. • Provides a comprehensive explanation of why reaction conditions are monitored or managed, relating these to the nature of the reaction • Provides an evaluation	6–7
• Demonstrates a thorough knowledge of the Haber process • Explains why some reaction conditions are monitored or managed	4–5
• Demonstrates a sound knowledge of the Haber process OR • Demonstrates a limited knowledge of the Haber process and identifies a reaction condition(s) that is (2 marks)/are monitored or managed (3 marks)	2–3
• Identifies the reactants and product of the Haber process OR • Gives one correct statement about the Haber process OR • Identifies one reaction condition that is monitored	1

Question 30 (a) (i)

Outcomes assessed: H6

MARKING GUIDELINES

Criteria	Marks
• Identifies one use of sulfuric acid in industry	1

Question 30 (a) (ii)

Outcomes assessed: H8

MARKING GUIDELINES

Criteria	Marks
- Provides full description of the extraction of sulfur as sulfur dioxide by the roasting of mineral sulfides AND - Includes a balanced equation for the process - Provides a complete description of the Frasch process for extracting sulfur from mineral deposits. (Mentions underground deposits of elemental sulfur, melting and mobilising it with superheated water and air, piping it to the surface)	3
- Provides a partial description the Frasch process OR - Correctly describes the extraction of sulfur as sulfur dioxide by roasting of mineral sulfides	2
- Describes one feature of the Frasch process OR - Identifies sulfide ores as a source of sulfur	1

Question 30 (b) (i)

Outcomes assessed: H8, H12, H13

MARKING GUIDELINES

Criteria	Marks
- Describes one precaution that could minimise hazards OR - Describes one precaution for the safe disposal of reactants or products	1

Question 30 (b) (ii)

Outcomes assessed: H8, H12, H13

MARKING GUIDELINES

Criteria	Marks
- Provides a complete sequence of steps in the investigation, including a description of techniques to identify at least two of the products OR - Correctly identifies three products AND gives the procedure used to identify each product	3
- Provides sequence of steps in the investigation AND - Shows how one product is correctly identified OR - Correctly names two products and outlines how each was identified	2
- Identifies a correct aspect of the procedure for this investigation OR - Correctly names one product and outlines how it was identified	1

Question 30 (c)

Outcomes assessed: H1, H4, H9, H13

MARKING GUIDELINES

Criteria	Marks
- Displays thorough knowledge of the structure of soaps and synthetic detergents - Displays thorough knowledge of the cleaning action of soaps and synthetic detergents - Makes an analysis linking understanding to development	4–5
- Displays knowledge of the structure of soaps and synthetic detergents OR - Displays knowledge of the cleaning action of soaps and synthetic detergents	2–3
- Provides a correct chemical name or chemical formula of a soap or synthetic detergent OR - Displays limited knowledge of the cleaning action of soaps and synthetic detergents OR - Identifies that soaps form insoluble precipitates with Ca and Mg ions OR - Correctly identifies the structures of soaps or detergents	1

Question 30 (d) (i)

Outcomes assessed: H8

MARKING GUIDELINES

Criteria	Marks
Identifies that only temperature changes affect the value of an equilibrium constant	1

Question 30 (d) (ii)

Outcomes assessed: H8

MARKING GUIDELINES

Criteria	Marks
- Describes any TWO of the following three: - An increase in pressure would increase the yield of NO₂ - Lowering the temperature would increase the yield of NO₂ - Selective removal/condensation of NO₂ from the mixture will increase its yield	2
Describes any one of the above three points	1

Question 30 (d) (iii)

Outcomes assessed: H8, H10, H12

MARKING GUIDELINES

Criteria	Marks
Gives the correct answer for the equilibrium constant, showing working	3
- Gives correct equilibrium concentration for gases OR - Correct working based on any equilibrium values - Gives correct equilibrium constant expression	2
- Correct working based on any equilibrium values OR - Gives correct equilibrium constant expression	1

Question 30 (e)

Outcomes assessed: H4, H8, H13

MARKING GUIDELINES

Criteria	Marks
- Displays extensive knowledge about an industrial method of preparation of BOTH an acid and a base AND provides a assessment of how environmental issues are addressed for each OR - Displays sound knowledge about an industrial method of BOTH and acid and a base, and assesses how at least two environmental issues are addressed for each AND - Includes correct chemical equations in answer	5–6
- Displays a knowledge about an industrial method of preparation of BOTH an acid and a base, and describes how at least two environmental issues are addressed OR - Displays sound knowledge about an industrial method of preparation of BOTH an acid and a base, and describes how one environmental issue is addressed for both methods	3–4
- Outlines an industrial method of preparation of either an acid or a base OR - Outlines how one environmental issue associated with an industrial method of preparation of either an acid or a base is addressed OR - Identifies an environmental issue associated with an industrial method of preparation of BOTH an acid and a base	2
- Identifies an industrial method of preparation of either an acid or a base OR - Identifies an environmental issue associated with an industrial method of preparation of either an acid or a base	1

Question 31 (a) (i)
Outcomes assessed: H8
MARKING GUIDELINES

Criteria	Marks
Correctly identifies ONE passivating metal	1

Question 31 (a) (ii)
Outcomes assessed: H6, H8, H13
MARKING GUIDELINES

Criteria	Marks
- Provides a sound explanation for the differences in corrosion of active and passivating metals - Includes a relevant balanced chemical equation	3
- Provides a sound explanation of the corrosion of EITHER active or passivating metals OR - Describes basic features of the corrosion of BOTH active and passivating metals OR - Describes basic features of the corrosion of EITHER active or passivating metals - Includes a relevant balanced chemical equation	2
Identifies basic features of the corrosion of EITHER an active metal or a passivating metal	1

Question 31 (b) (i)
Outcomes assessed: H8, H12, H13
MARKING GUIDELINES

Criteria	Marks
- Describes ONE precaution that could minimise hazards OR - Describes ONE precaution for the safe disposal of reactants or products	1

Question 31 (b) (ii)

Outcomes assessed: H8, H12, H13

MARKING GUIDELINES

Criteria	Marks
- Provides a complete sequence of steps in the investigation - States that other variables are controlled - Identifying how the observed variable is measured	3
- Provides an incomplete sequence of steps in the investigation including identifying how the observed variable is measured - OR - Provides an incomplete sequence of steps in the investigation, including stating that other variables are controlled	2
Identifies a correct aspect of the procedure for an electrolysis investigation	1

Question 31 (c)

Outcomes assessed: H1, H2, H8, H13

MARKING GUIDELINES

Criteria	Marks
- Demonstrates a thorough knowledge of the factors that affect corrosion in deep water - Includes relevant equations (sulfate reduction; water and oxygen) - Analyses how theories have changed	4–5
- Demonstrates sound knowledge of more than one factor that affects corrosion in deep water have changed AND - Compares two theories of corrosion	3
- Demonstrates a sound knowledge of ONE factor that affects corrosion in deep water OR - Demonstrates a limited knowledge of how theories about corrosion at great ocean depth have changed	2
Identifies ONE factor that affects the rate of corrosion	1

Question 31 (d) (i)*Outcomes assessed: H8***MARKING GUIDELINES**

Criteria	Marks
• Correct identifies the metal	1

Question 31 (d) (ii)*Outcomes assessed: H4, H8, H13, H14***MARKING GUIDELINES**

Criteria	Marks
• Provides a thorough description of the effectiveness of each container in limiting corrosion, and provides a comparison	4–5
• Provides a thorough description of corrosion in ONE container AND • Provides a limited description of corrosion in TWO containers	2–3
• Provides a limited description of corrosion in ONE container	1

Question 31 (e)

Outcomes assessed: H3, H7, H8, H13

MARKING GUIDELINES

Criteria	Marks
- Displays a thorough knowledge of electrolysis - Provides an assessment of how an understanding of electrolysis has contributed to the development of methods for restoring and cleaning marine artefacts	5–6
- Displays sound knowledge of two or more of: electrolysis, corrosion of metallic objects, methods of restoration and cleaning of marine artefacts - Provides an assessment of how an understanding of electrolysis has contributed to the development of methods for restoring and cleaning marine artefacts	4
Displays sound knowledge of two or more of: electrolysis, corrosion of metallic objects, methods of restoration and cleaning of marine artefacts	3
Displays sound knowledge of one of: electrolysis, corrosion of metallic objects, methods of restoration and cleaning of marine artefacts	2
- Displays limited knowledge of electrolysis OR - Displays limited knowledge of corrosion of metallic objects OR - Displays limited knowledge of methods of restoring and cleaning marine artefacts	1

Question 32 (a) (i)

Outcomes assessed: H4, H9

MARKING GUIDELINES

Criteria	Marks
Identifies the role of glycogen molecules in human muscle and liver	1

Question 32 (a) (ii)

Outcomes assessed: H6, H9, H13

MARKING GUIDELINES

Criteria	Marks
- Demonstrates knowledge of bond formation between glucose molecules to give glycogen, including: - identifying that it involves a condensation reaction between hydroxyl groups on different glucose molecules - identifying and describing the glycosidic bond produced OR - Fully labelled structured equation	3
- Demonstrates knowledge of bond formation between glucose molecules (diagrams acceptable) to give glycogen, including: - identifying that it involves a condensation reaction between hydroxyl groups on different glucose monomers OR - Demonstrates knowledge of bond formation between glucose monomers and that it is a glycosidic bond that is produced	2
- Identifies that a condensation reaction is involved OR - Identifies that water is produced OR - Identifies that a glycosidic bond is produced	1

Question 32 (b) (i)

Outcomes assessed: H2, H9, H12, H13

MARKING GUIDELINES

Criteria	Marks
Describes the structures of fatty acids and glycerol including a difference or similarity between structures	2
Describes the structure of either fatty acids OR glycerol	1

Question 32 (b) (ii)

Outcomes assessed: H9

MARKING GUIDELINES

Criteria	Marks
- Identifies that glycerol is polar because it is a smaller molecule and has three hydroxyl groups - Identifies that fatty acids are not polar because of their long hydrocarbon chain	2
- Identifies that glycerol is polar because it is small and has three polar hydroxyl groups OR - Identifies that fatty acids that are commonly found in the body are non-polar because of their long hydrocarbon chain OR - Identifies that the solubility of a substance in water increases with its polarity	1

Question 32 (c)

Outcomes assessed: H1, H7, H8, H9, H13

MARKING GUIDELINES

Criteria	Marks
- Displays thorough knowledge about protein composition and structure - Displays thorough knowledge about the theory of muscle contraction - Makes an analysis linking understanding to the current theory	4–5
- Displays knowledge about protein composition and structure OR - Displays knowledge the sliding filament theory of muscle contraction	2–3
- Identifies one fact about protein composition or structure OR - Identifies one or more proteins involved in muscle contraction OR - Identifies the sliding filament theory of muscle contraction	1

Question 32 (d) (i)

Outcomes assessed: H7, H9

MARKING GUIDELINES

Criteria	Marks
Correctly identifies glycolysis	1

Question 32 (d) (ii)

Outcomes assessed: H7, H9, H10

MARKING GUIDELINES

Criteria	Marks
Constructs an equation that shows that 2 molecules of ATP and 2 molecules of pyruvate are produced from one molecule of glucose, two molecules of ADP and two molecules of phosphate	2
OR 1 glucose + 2 ADP + 2 phosphate → 2 pyruvate + 2 ATP	
Constructs an equation that identifies both ATP AND pyruvate as the products of glycolysis	1

Question 32 (d) (iii)

Outcomes assessed: H2, H8, H9, H10, H12, H13

MARKING GUIDELINES

Criteria	Marks
- Compares the number of ATP produced per carbon atom in fatty acids and in carbohydrates - Identifies one associated pathway OR - Relates the difference in number of ATP molecules produced to the oxidation level of carbon in fatty acids and carbohydrates	2–3
- Calculates that 8.1ATP are produced per carbon of stearic acid and 6 ATP are produced per carbon of glucose OR Identifies that: - Fats (fatty acids) release more energy when they are oxidised than glucose OR - Carbon atoms in fatty acids exist in a lower oxidation state OR - Fatty acids require 2ATP to produce 148 ATP OR - Glucose requires 2ATP to produce 38 ATP OR - Explanation of diagram	1

Question 32 (e)

Outcomes assessed: H1, H7, H8, H9, H10

MARKING GUIDELINES

Criteria	Marks
- Demonstrates extensive knowledge of two or more metabolic pathways (processes) and of the chemistry of ATP - Provides an evaluation, linking the importance of the chemistry of ATP to metabolic processes	5–6
- Demonstrates sound knowledge of two or more metabolic pathways (processes) and of the chemistry of ATP OR - Demonstrates a sound knowledge of one metabolic pathway (process) AND evaluates the importance of the chemistry of ATP	3–4
- Identifies two or more metabolic processes (or pathways) OR - Identifies a metabolic pathway (or process) and that ATP acts as an energy store in the body	2
- Identifies a metabolic pathway (or process) OR - Identifies that ATP acts as an energy store in the body OR - Identifies that ATP stands for adenosine triphosphate	1

Question 33 (a) (i)

Outcomes assessed: H1, H6

MARKING GUIDELINES

Criteria	Marks
Correctly identifies the maximum number of electrons	1

Question 33 (a) (ii)

Outcomes assessed: H7

MARKING GUIDELINES

Criteria	Marks
Provides a complete explanation of how data can be used to provide information about the arrangement of electrons around the atoms	3
Provides a partial explanation of how data can be used to provide information about the arrangement of electrons around the atoms	2
- Identifies the number of electrons in the outer shell correctly for Na, Al and Mg OR - Identifies that ionisation energy increases as the number of electrons removed increases OR - Identifies that ionisation energy increases sharply as an electron is removed from an inner shell	1

Question 33 (b) (i)

Outcomes assessed: H8, H12, H13

MARKING GUIDELINES

Criteria	Marks
Explains two precautions correctly	2
- Explains one precaution OR - States two precautions	1

Question 33 (b) (ii)

Outcomes assessed: H8, H12, H13

MARKING GUIDELINES

Criteria	Marks
Provides a complete sequence of steps in the investigation	2
Identifies a correct aspect of the procedure for this investigation	1

Question 33 (c)

Outcomes assessed: H1, H2, H4

MARKING GUIDELINES

Criteria	Marks
Demonstrates a thorough knowledge of the relationship between properties and chemical composition of pigments used in traditional art by Aboriginal people	4–5
Demonstrates a sound knowledge of the relationship between properties and chemical composition for at least one named pigment used in traditional art by Aboriginal people	2–3
- Names a pigment and gives its chemical composition correctly OR - Gives the correct colour and chemical composition for a pigment (not named) OR - Gives a brief description of a property (other than colour) OR - Names TWO pigments without their chemical composition	1

Question 33 (d) (i)

Outcomes assessed: H6

MARKING GUIDELINES

Criteria	Marks
Correctly identifies the block in which mercury is present	1

Question 33 (d) (ii)

Outcomes assessed: H6, H7, H8

MARKING GUIDELINES

Criteria	Marks
Provides a complete explanation of why transition metals have more than one oxidation state	2
Provides a partial explanation of why transition metals have more than one oxidation state	1

Question 33 (d) (iii)

Outcomes assessed: H3, H4, H6, H8, H13

MARKING GUIDELINES

Criteria	Marks
• Explains why chelating ligands are used to treat poisoning by heavy metals	2–3
• Identifies one correct aspect of ligand bonding	1

Question 33 (e)

Outcomes assessed: H1, H4, H6, H7, H13

MARKING GUIDELINES

Criteria	Marks
• Demonstrates an extensive knowledge of the range of technologies	5–6
• Provides an evaluation of these technologies	
• Describes two or more technologies and how they are used to identify pigments	3–4
• Describes a technology used to identify pigments	2
OR	
• Identifies two or more technologies used to identify pigments	1
• Identifies one technology used to identify pigments	

Question 34 (a) (i)

Outcomes assessed: H9, H10

MARKING GUIDELINES

Criteria	Marks
• Correctly identifies the class of compounds	1

Question 34 (a) (ii)

Outcomes assessed: H8, H9

MARKING GUIDELINES

Criteria	Marks
• Describes two different tests. The second test must demonstrate different reagents and conditions	3
• Describes one test	2
• Outlines a second test for a separate class of compounds	
• Identifies one distinguishing test for a specific class	1

Question 34 (b) (i)

Outcomes assessed: H12, H13

MARKING GUIDELINES

Criteria	Marks
- Describes one precaution that could minimise hazards for this specific test OR - Describes one precaution for the safe disposal of reactants or products for this specific test	1

Question 34 (b) (ii)

Outcomes assessed: H12, H13

MARKING GUIDELINES

Criteria	Marks
Provides a complete sequence of appropriate steps in the investigation, including class of sugars which has the correct colour change (blue to orange)	3
Provides an incomplete sequence of steps in the investigation	2
Identifies a correct aspect of the procedure for this investigation	1

Question 34 (c)

Outcomes assessed: H3, H4, H6, H7, H13

MARKING GUIDELINES

Criteria	Marks
Demonstrates a thorough knowledge of emission spectra of elements and how they can assist in the identification of the origins of a mixture	4–5
Demonstrates a sound knowledge of emission spectra and how they are used to identify elements	2–3
Identifies a feature of emission spectra	1

Question 34 (d) (i)

Outcomes assessed: H4, H12, H14

MARKING GUIDELINES

Criteria	Marks
Correctly identifies the more probable father	1

Question 34 (d) (ii)

Outcomes assessed: H2, H9, H10, H13

MARKING GUIDELINES

Criteria	Marks
• Outlines the structure (helical strands/connection between helices; at complementary N bases) and composition (sugar groups, phosphate, four different N bases) of DNA	2
• Outline the structure of DNA OR • Outlines the composition of DNA OR • Basic knowledge of both structure and composition of DNA	1

Question 34 (d) (iii)

Outcomes assessed: H1, H3, H4, H5, H13

MARKING GUIDELINES

Criteria	Marks
• Provides a thorough description of how DNA analysis can be used to show that two people belong to the same family by providing at least three steps in DNA analysis (including PCR, electrophoresis and introns, isolation of DNA...)	2–3
• Outlines one relevant aspect of DNA analysis OR • Outlines the similarities in DNA material amongst family members compared with non-related people	1

Question 34 (e)

Outcomes assessed: H1, H3, H4, H8, H9, H13

MARKING GUIDELINES

Criteria	Marks
• Demonstrates extensive knowledge of how chromatographic methods have advanced forensic science	5–6
• Describes two or more chromatographic methods	3–4
• Describes a chromatographic method OR • Identifies two or more chromatographic methods	2
• Identifies one chromatographic method	1